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### Heterocyclic compound and organic light emitting device comprising same

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#### Abstract

The present specification relates to a heterocyclic compound represented by Chemical Formula 1, and an organic light emitting device including the same.

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## Background/Summary

### TECHNICAL FIELD

(1) The present specification relates to a heterocyclic compound, and an organic light emitting device including the same.

(2) The present specification claims priority to and the benefits of Korean Patent Application No. 10-2019-0088238, filed with the Korean Intellectual Property Office on Jul. 22, 2019, the entire contents of which are incorporated herein by reference.

### BACKGROUND ART

(3) An electroluminescent device is one type of self-emissive display devices, and has an advantage of having a wide viewing angle, and a high response speed as well as having an excellent contrast.

(4) An organic light emitting device has a structure disposing an organic thin film between two electrodes. When a voltage is applied to an organic light emitting device having such a structure, electrons and holes injected from the two electrodes bind and pair in the organic thin film, and light emits as these annihilate. The organic thin film may be formed in a single layer or a multilayer as necessary.

(5) A material of the organic thin film may have a light emitting function as necessary. For example, as a material of the organic thin film, compounds capable of forming a light emitting layer themselves alone may be used, or compounds capable of performing a role of a host or a dopant of a host-dopant-based light emitting layer may also be used. In addition thereto, compounds capable of performing roles of hole injection, hole transfer, electron blocking, hole blocking, electron transfer, electron injection and the like may also be used as a material of the organic thin film.

(6) Development of an organic thin film material has been continuously required for enhancing performance, lifetime or efficiency of an organic light emitting device.

### DISCLOSURE

#### Technical Problem

(7) The present specification is directed to providing a heterocyclic compound, and an organic light emitting device including the same.

#### Technical Solution

(8) One embodiment of the present specification provides a heterocyclic compound represented by the following Chemical Formula 1.

(9) ##STR00001##

(10) In Chemical Formula 1, X is O or S, L is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, Z and R1 are each independently a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2 to C60 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group, R2 to R7 are each independently hydrogen; deuterium; a halogen group; a cyano group; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, a is 1 or 2, b is an integer of 1 to 3, m and n are each an integer of 1 to 5, and when a, b, m and n are each 2 or greater, substituents in the parentheses are the same as or different from each other.

(11) Another embodiment of the present application provides an organic light emitting device including a first electrode; a second electrode provided opposite to the first electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes the heterocyclic compound represented by Chemical Formula 1.

(12) Another embodiment of the present application provides an organic light emitting device including a first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and a second electrode provided on the second stack, wherein the charge generation layer includes the heterocyclic compound represented by Chemical Formula 1.

#### Advantageous Effects

(13) A compound described in the present specification can be used as a material of an organic material layer of an organic light emitting device. In the organic light emitting device, the compound is capable of performing a role of a hole injection material, a hole transfer material, a light emitting material, an electron transfer material, an electron injection material or the like. Particularly, the compound can be used as an electron transfer layer material or a charge generation layer material of an organic light emitting device.

(14) Particularly, Chemical Formula 1 has a structure in which dibenzofuran or dibenzothiophene is fused to quinoline, and HOMO and LUMO levels can be adjusted by having various substituents in the benzene ring which is not fused to the quinoline in the dibenzofuran or dibenzothiophene. Accordingly, electrical properties of the structure can be enhanced by adjusting an energy band gap. In addition, electron transfer capability of a material can be enhanced through substituents having strong electron migration properties.

(15) Specifically, when using the compound represented by Chemical Formula 1 in an organic material layer, a driving voltage of the device can be lowered, light efficiency can be enhanced, and lifetime properties of the device can be enhanced.

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## Description

### DESCRIPTION OF DRAWINGS

(1) FIG. 1 to FIG. 4 each illustrate a lamination structure of an organic light emitting device according to one embodiment of the present specification.

(2) **100**: Substrate **200**: Anode **300**: Organic Material Layer **301**: Hole Injection Layer **302**: Hole Transfer Layer **303**: Light Emitting Layer **304**: Electron Transfer Layer **305**: Electron Injection Layer **400**: Cathode

### MODE FOR DISCLOSURE

(3) Hereinafter, the present specification will be described in more detail.

(4) In the present specification, a certain part “including” certain constituents means capable of further including other constituents, and does not exclude other constituents unless particularly stated on the contrary.

(5) In the present specification, the term “substitution” means a hydrogen atom bonding to a carbon atom of a compound being changed to another substituent, and the position of substitution is not limited as long as it is a position at which the hydrogen atom is substituted, that is, a position at which a substituent can substitute, and when two or more substituents substitute, the two or more substituents may be the same as or different from each other.

(6) In the present specification, a “T1 value” means an energy level value in a triplet state.

(7) In the present specification, “substituted or unsubstituted” means being substituted with one or more substituents selected from the group consisting of a halogen group; a cyano group; a C1 to C60 linear or branched alkyl group; a C2 to C60 linear or branched alkenyl group; a C2 to C60

linear or branched alkynyl group; a C3 to C60 monocyclic or polycyclic cycloalkyl group; a C2 to C60 monocyclic or polycyclic heterocycloalkyl group; a C6 to C60 monocyclic or polycyclic aryl group; a C2 to C60 monocyclic or polycyclic heteroaryl group; a silyl group; a phosphine oxide group; and an amine group, or being unsubstituted, or being substituted with a substituent linking two or more substituents selected from among the substituents illustrated above, or being unsubstituted.

(8) In the present specification, the halogen may be fluorine, chlorine, bromine or iodine.

(9) In the present specification, the alkyl group includes linear or branched having 1 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkyl group may be from 1 to 60, specifically from 1 to 40 and more specifically from 1 to 20. Specific examples thereof may include a methyl group, an ethyl group, a propyl group, an n-propyl group, an isopropyl group, a butyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a sec-butyl group, a 1-methyl-butyl group, a 1-ethylbutyl group, a pentyl group, an n-pentyl group, an isopentyl group, a neopentyl group, a tert-pentyl group, a hexyl group, an n-hexyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 4-methyl-2-pentyl group, a 3,3-dimethylbutyl group, a 2-ethylbutyl group, a heptyl group, an n-heptyl group, a 1-methylhexyl group, a cyclopentylmethyl group, a cyclohexylmethyl group, an octyl group, an n-octyl group, a tert-octyl group, a 1-methylheptyl group, a 2-ethylhexyl group, a 2-propylpentyl group, an n-nonyl group, a 2,2-dimethylheptyl group, a 1-ethyl-propyl group, a 1,1-dimethyl-propyl group, an isohexyl group, a 2-methylpentyl group, a 4-methylhexyl group, a 5-methylhexyl group and the like, but are not limited thereto.

(10) In the present specification, the alkenyl group includes linear or branched having 2 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkenyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 2 to 20. Specific examples thereof may include a vinyl group, a 1-propenyl group, an isopropenyl group, a 1-butenyl group, a 2-butenyl group, a 3-butenyl group, a 1-pentenyl group, a 2-pentenyl group, a 3-pentenyl group, a 3-methyl-1-butenyl group, a 1,3-butadienyl group, an allyl group, a 1-phenylvinyl-1-yl group, a 2-phenylvinyl-1-yl group, a 2,2-diphenylvinyl-1-yl group, a 2-phenyl-2-(naphthyl-1-yl)vinyl-1-yl group, a 2,2-bis(diphenyl-1-yl)vinyl-1-yl group, a stilbenyl group, a styrenyl group and the like, but are not limited thereto.

(11) In the present specification, the alkynyl group includes linear or branched having 2 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkynyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 2 to 20.

(12) In the present specification, the cycloalkyl group includes monocyclic or polycyclic having 3 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the cycloalkyl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a cycloalkyl group, but may also be different types of cyclic groups such as a heterocycloalkyl group, an aryl group and a heteroaryl group. The number of carbon groups of the cycloalkyl group may be from 3 to 60, specifically from 3 to 40 and more specifically from 5 to 20. Specific examples thereof may include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a 3-methylcyclopentyl group, a 2,3-dimethylcyclopentyl group, a cyclohexyl group, a 3-methylcyclohexyl group, a 4-methylcyclohexyl group, a 2,3-dimethylcyclohexyl group, a 3,4,5-trimethylcyclohexyl group, a 4-tert-butylcyclohexyl group, a cycloheptyl group, a cyclooctyl group and the like, but are not limited thereto.

(13) In the present specification, the heterocycloalkyl group includes O, S, Se, N or Si as a heteroatom, includes monocyclic or polycyclic having 2 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the heterocycloalkyl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a heterocycloalkyl group, but may also be different types of cyclic groups

such as a cycloalkyl group, an aryl group and a heteroaryl group. The number of carbon atoms of the heterocycloalkyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 3 to 20.

(14) In the present specification, the aryl group includes monocyclic or polycyclic having 6 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the aryl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be an aryl group, but may also be different types of cyclic groups such as a cycloalkyl group, a heterocycloalkyl group and a heteroaryl group. The aryl group includes a spiro group. The number of carbon atoms of the aryl group may be from 6 to 60, specifically from 6 to 40 and more specifically from 6 to 25. Specific examples of the aryl group may include a phenyl group, a biphenyl group, a triphenyl group, a naphthyl group, an anthryl group, a chrysenyl group, a phenanthrenyl group, a perylenyl group, a fluoranthenyl group, a triphenylenyl group, a phenalenyl group, a pyrenyl group, a tetracenyl group, a pentacenyl group, a fluorenyl group, an indenyl group, an acenaphthylenyl group, a benzofluorenyl group, a spirobifluorenyl group, a 2,3-dihydro-1H-indenyl group, a fused ring thereof, and the like, but are not limited thereto.

(15) In the present specification, the silyl group is a substituent including Si, having the Si atom directly linked as a radical, and is represented by —SiR.sub.104R.sub.105R.sub.106. R.sub.104 to R.sub.106 are the same as or different from each other, and may be each independently a substituent formed with at least one of hydrogen; deuterium; a halogen group; an alkyl group; an alkenyl group; a cycloalkyl group; an aryl group; and a heterocyclic group. Specific examples of the silyl group may include a trimethylsilyl group, a triethylsilyl group, a t-butyl dimethylsilyl group, a vinyl dimethylsilyl group, a propyl dimethylsilyl group, a triphenylsilyl group, a diphenylsilyl group, a phenylsilyl group and the like, but are not limited thereto.

(16) In the present specification, the fluorenyl group may be substituted, and adjacent substituents may bond to each other to form a ring.

(17) When the fluorenyl group is substituted,

(18) ##STR00002##

and the like may be included, however, the structure is not limited thereto.

(19) In the present specification, the heteroaryl group includes O, S, Se, N or Si as a heteroatom, includes monocyclic or polycyclic having 2 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the heteroaryl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a heteroaryl group, but may also be different types of cyclic groups such as a cycloalkyl group, a heterocycloalkyl group and an aryl group. The number of carbon atoms of the heteroaryl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 3 to 25. Specific examples of the heteroaryl group may include a pyridyl group, a pyrrolyl group, a pyrimidyl group, a pyridazinyl group, a furanyl group, a thiophene group, an imidazolyl group, a pyrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, a triazolyl group, a furazanyl group, an oxadiazolyl group, a thiadiazolyl group, a dithiazolyl group, a tetrazolyl group, a pyranyl group, a thiopyranyl group, a diazinyl group, an oxazinyl group, a thiazinyl group, a dioxynyl group, a triazinyl group, a tetrazinyl group, a quinolyl group, an isoquinolyl group, a quinazolinyl group, an isoquinazolinyl group, a quinoxalinyl group, a naphthyridyl group, an acridinyl group, a phenanthridinyl group, an imidazopyridinyl group, a diazanaphthalenyl group, a triazaindene group, an indolyl group, an indolizinyl group, a benzothiazolyl group, a benzoxazolyl group, a benzimidazolyl group, a benzothiophene group, a benzofuran group, a dibenzothiophene group, a dibenzofuran group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a phenazinyl group, a dibenzosilole group, spirobi(dibenzosilole), a dihydrophenazinyl group, a phenoxazinyl group, a phenanthridyl group, an imidazopyridinyl group, a thienyl group, an indolo[2,3-a]carbazolyl group, an indolo[2,3-b]carbazolyl group, an indolinyl group, a 10,11-dihydro-dibenzo[b,f]azepine group, a 9,10-dihydroacridinyl group, a phenanthrazinyl group, a

phenothiazinyl group, a phthalazinyl group, a naphthylidiny group, a phenanthrolinyl group, a benzo[c][1,2,5]thiadiazolyl group, a 5,10-dihydrobenzo[b,e][1,4]azasilinyl group, a pyrazolo[1,5-c]quinazolinyl group, a pyrido[1,2-b]indazolyl group, a pyrido[1,2-a]imidazo[1,2-e]indolinyl group, a 5,11-dihydroindeno[1,2-b]carbazolyl group and the like, but are not limited thereto.

(20) In the present specification, the phosphine oxide group may be specifically substituted with an aryl group, and examples described above may be applied to the aryl group. Examples of the phosphine oxide group may include a diphenylphosphine oxide group, a dinaphthylphosphine oxide group and the like, but are not limited thereto.

(21) In the present specification, the amine group may be selected from the group consisting of a monoalkylamine group; a monoarylamine group; a monoheteroarylamine group; —NH.sub.2; a dialkylamine group; a diarylamine group; a diheteroarylamine group; an alkylarylamine group; an alkylheteroarylamine group; and an arylheteroarylamine group, and although not particularly limited thereto, the number of carbon atoms is preferably from 1 to 30. Specific examples of the amine group may include a methylamine group, a dimethylamine group, an ethylamine group, a diethylamine group, a phenylamine group, a naphthylamine group, a biphenylamine group, a dibiphenylamine group, an anthracenylamine group, a 9-methyl-anthracenylamine group, a diphenylamine group, a phenylnaphthylamine group, a ditolylamine group, a phenyltolylamine group, a triphenylamine group, a biphenylnaphthylamine group, a phenylbiphenylamine group, a biphenylfluorenylamine group, a phenyltriphenylenylamine group, a biphenyltriphenylenylamine group and the like, but are not limited thereto.

(22) In the present specification, the structures illustrated as the aryl group and the heteroaryl group described above may be used as the arylene group and the heteroarylene group except that they are not a monovalent group.

(23) In the present specification, an “adjacent” group may mean a substituent substituting an atom directly linked to an atom substituted by the corresponding substituent, a substituent sterically most closely positioned to the corresponding substituent, or another substituent substituting an atom substituted by the corresponding substituent. For example, two substituents substituting ortho positions in a benzene ring, and two substituents substituting the same carbon in an aliphatic ring may be interpreted as groups “adjacent” to each other.

(24) As an aliphatic or aromatic hydrocarbon ring or heteroring that adjacent groups may form, the structures illustrated as the cycloalkyl group, the heterocycloalkyl group, the aryl group and the heteroaryl group described above may be applied except for those that are not a monovalent group.

(25) One embodiment of the present specification provides a heterocyclic compound represented by Chemical Formula 1.

(26) Particularly, Chemical Formula 1 has a structure in which dibenzofuran or dibenzothiophene is fused to quinoline, and HOMO and LUMO levels can be adjusted by having various substituents in the benzene ring which is not fused to the quinoline in the dibenzofuran or dibenzothiophene. Accordingly, electrical properties of the structure may be enhanced by adjusting an energy band gap. In addition, electron transfer capability of a material may be enhanced through substituents having strong electron migration properties.

(27) In one embodiment of the present specification, X is O or S.

(28) In one embodiment of the present specification, X is O.

(29) In another embodiment, X is S.

(30) In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group.

(31) By the Z substituent and the core structure being separated by L in the compound, hole migration properties of the core structure and electron migration properties of the substituent are distinguished in the electron cloud distribution of the molecule. This is an excellent bipolar structure facilitating electron migration, and therefore, efficiency, lifetime and driving voltage may

be significantly improved.

(32) In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted C6 to C30 arylene group; or a substituted or unsubstituted C2 to C30 heteroarylene group.

(33) In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted phenylene group; a substituted or unsubstituted biphenylene group; a substituted or unsubstituted terphenylene group; a substituted or unsubstituted naphthylene group; a substituted or unsubstituted anthracenylene group; a substituted or unsubstituted triphenylenylene group; a substituted or unsubstituted phenanthrenylene group; a substituted or unsubstituted divalent pyridine group; a substituted or unsubstituted divalent pyrimidine group; or a substituted or unsubstituted divalent triazine group.

(34) In one embodiment of the present specification, L is a direct bond; a phenylene group unsubstituted or substituted with an aryl group or a heteroaryl group; a biphenylene group; a terphenylene group; a naphthylene group; an anthracenylene group; a triphenylenylene group; a phenanthrenylene group; a divalent pyridine group unsubstituted or substituted with an aryl group or a heteroaryl group; a divalent pyrimidine group unsubstituted or substituted with an aryl group; or a divalent triazine group unsubstituted or substituted with an aryl group or a heteroaryl group.

(35) In one embodiment of the present specification, L is a direct bond; a phenylene group unsubstituted or substituted with a phenyl group or a carbazole group; a biphenylene group; a terphenylene group; a naphthylene group; an anthracenylene group; a triphenylenylene group; a phenanthrenylene group; a divalent pyridine group unsubstituted or substituted with a phenyl group or a pyridine group; a divalent pyrimidine group unsubstituted or substituted with a phenyl group or a biphenyl group; or a divalent triazine group unsubstituted or substituted with a phenyl group unsubstituted or substituted with a cyano group, a biphenyl group, a naphthyl group, or a carbazole group unsubstituted or substituted with a phenyl group.

(36) In one embodiment of the present specification, Z is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2 to C60 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

(37) In one embodiment of the present specification, Z is a substituted or unsubstituted C1 to C30 alkyl group; a substituted or unsubstituted C2 to C30 alkenyl group; a substituted or unsubstituted C2 to C30 alkynyl group; a substituted or unsubstituted C3 to C30 cycloalkyl group; a substituted or unsubstituted C2 to C30 heterocycloalkyl group; a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group; a substituted or unsubstituted C1 to C10 alkylamine group; a substituted or unsubstituted C6 to C30 arylamine group; a substituted or unsubstituted C2 to C30 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

(38) In one embodiment of the present specification, Z is a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group; or a substituted or unsubstituted phosphine oxide group.

(39) In one embodiment of the present specification, Z is a substituted or unsubstituted phenyl group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a substituted or unsubstituted pyridine group; a substituted or unsubstituted pyrimidine group; a substituted or unsubstituted triazine group; a

substituted or unsubstituted carbazole group; a substituted or unsubstituted benzocarbazole group; a substituted or unsubstituted phenanthrolinyl group; a substituted or unsubstituted dibenzofuran group; a substituted or unsubstituted dibenzothiophene group; a substituted or unsubstituted dihydroacridine group; or a substituted or unsubstituted phosphine oxide group.

(40) In one embodiment of the present specification, Z is a phenyl group unsubstituted or substituted with a cyano group, a phenyl group or a carbazole group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a pyridine group unsubstituted or substituted with a phenyl group or a pyridine group; a pyrimidine group unsubstituted or substituted with a phenyl group or a biphenyl group; a triazine group unsubstituted or substituted with a phenyl group unsubstituted or substituted with a naphthyl group or a cyano group, a biphenyl group, a naphthyl group, a dimethylfluorenyl group, a dibenzofuran group, a dibenzothiophene group, or a carbazole group unsubstituted or substituted with a phenyl group; a carbazole group unsubstituted or substituted with a phenyl group; a benzocarbazole group; a phenanthrolinyl group unsubstituted or substituted with a phenyl group; a dibenzofuran group unsubstituted or substituted with a phenyl group; a dibenzothiophene group unsubstituted or substituted with a phenyl group; a dihydroacridine group unsubstituted or substituted with a methyl group or a phenyl group; or a phosphine oxide group unsubstituted or substituted with a phenyl group.

(41) In one embodiment of the present specification, R1 is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2 to C60 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

(42) In one embodiment of the present specification, R1 is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(43) In one embodiment of the present specification, R1 is a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(44) In one embodiment of the present specification, R1 is a substituted or unsubstituted C6 to C30 aryl group; or a substituted or unsubstituted C2 to C30 heteroaryl group.

(45) In one embodiment of the present specification, R1 is a substituted or unsubstituted C6 to C30 aryl group.

(46) In one embodiment of the present specification, R1 is a substituted or unsubstituted phenyl group; a substituted or unsubstituted naphthyl group; a substituted or unsubstituted anthracenyl group; a substituted or unsubstituted phenanthrenyl group; a substituted or unsubstituted pyrenyl group; or a substituted or unsubstituted perylene group.

(47) In one embodiment of the present specification, R1 is a phenyl group; a naphthyl group; an anthracenyl group; a phenanthrenyl group; a pyrenyl group; or a perylene group.

(48) In one embodiment of the present specification, R2 to R7 are each independently hydrogen; deuterium; a halogen group; a cyano group; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(49) In one embodiment of the present specification, R2 to R7 are each independently hydrogen; deuterium; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6

to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(50) In one embodiment of the present specification, R2 to R7 are each independently hydrogen; or deuterium.

(51) In one embodiment of the present specification, R2 to R7 are hydrogen.

(52) In one embodiment of the present specification, m may be 1 or 2.

(53) In one embodiment of the present specification, n may be an integer of 1 to 5.

(54) In one embodiment of the present specification, Chemical Formula 1 is represented by the following Chemical Formula 1-1.

(55) ##STR00003##

(56) In Chemical Formula 1-1, X, L, Z, R2 to R7, a, b, m and n have the same definitions as in Chemical Formula 1, L' is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, Z' is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(57) In one embodiment of the present specification, L' is a direct bond.

(58) In one embodiment of the present specification, Z' is a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(59) In one embodiment of the present specification, Z' is a substituted or unsubstituted C6 to C60 aryl group.

(60) In one embodiment of the present specification, Z' is a substituted or unsubstituted C6 to C30 aryl group.

(61) In one embodiment of the present specification, Z' is a phenyl group; a naphthyl group; an anthracenyl group; a phenanthrenyl group; a pyrenyl group; or a perylene group.

(62) In the heterocyclic compound provided in one embodiment of the present specification, Chemical Formula 1 is represented by any one of the following Chemical Formulae 1-2 to 1-5.

(63) ##STR00004##

(64) In Chemical Formulae 1-2 to 1-5, X, L, Z, R1 to R7, a, b, m and n have the same definitions as in Chemical Formula 1.

(65) In the heterocyclic compound provided in one embodiment of the present specification, Chemical Formula 1 is represented by any one of the following Chemical Formulae 2 to 4.

(66) ##STR00005##

(67) In Chemical Formulae 2 to 4, X, L, Z, R1 to R7, a, b, m and n have the same definitions as in Chemical Formula 1.

(68) In one embodiment of the present specification, Chemical Formula 1 may be represented by any one of the following compounds, but is not limited thereto.

(69) ##STR00006## ##STR00007## ##STR00008## ##STR00009## ##STR00010##

##STR00011## ##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016##

##STR00017## ##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022##

##STR00023## ##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028##

##STR00029## ##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034##

##STR00035## ##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040##

##STR00041## ##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046##

##STR00047## ##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052##

##STR00053## ##STR00054## ##STR00055## ##STR00056## ##STR00057## ##STR00058##

##STR00059## ##STR00060## ##STR00061## ##STR00062## ##STR00063## ##STR00064##

##STR00065## ##STR00066## ##STR00067## ##STR00068## ##STR00069## ##STR00070##

##STR00071## ##STR00072## ##STR00073## ##STR00074## ##STR00075## ##STR00076##

##STR00077## ##STR00078## ##STR00079## ##STR00080## ##STR00081## ##STR00082##

##STR00083## ##STR00084## ##STR00085## ##STR00086## ##STR00087## ##STR00088##

##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093## ##STR00094##  
(70) ##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099##  
##STR00100## ##STR00101## ##STR00102## ##STR00103## ##STR00104## ##STR00105##  
##STR00106## ##STR00107## ##STR00108## ##STR00109## ##STR00110## ##STR00111##  
##STR00112## ##STR00113## ##STR00114## ##STR00115## ##STR00116## ##STR00117##  
##STR00118## ##STR00119## ##STR00120## ##STR00121## ##STR00122## ##STR00123##  
##STR00124## ##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129##  
##STR00130## ##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135##  
##STR00136## ##STR00137## ##STR00138## ##STR00139## ##STR00140## ##STR00141##  
##STR00142## ##STR00143## ##STR00144## ##STR00145## ##STR00146## ##STR00147##  
##STR00148## ##STR00149## ##STR00150## ##STR00151## ##STR00152## ##STR00153##  
##STR00154## ##STR00155## ##STR00156## ##STR00157## ##STR00158## ##STR00159##  
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##STR00166## ##STR00167## ##STR00168## ##STR00169## ##STR00170## ##STR00171##  
##STR00172## ##STR00173## ##STR00174## ##STR00175## ##STR00176## ##STR00177##  
##STR00178## ##STR00179## ##STR00180## ##STR00181## ##STR00182## ##STR00183##  
##STR00184## ##STR00185## ##STR00186## ##STR00187## ##STR00188## ##STR00189##  
##STR00190## ##STR00191## ##STR00192## ##STR00193## ##STR00194## ##STR00195##  
(71) ##STR00196## ##STR00197## ##STR00198## ##STR00199## ##STR00200##  
##STR00201## ##STR00202## ##STR00203## ##STR00204## ##STR00205## ##STR00206##  
##STR00207## ##STR00208## ##STR00209## ##STR00210## ##STR00211## ##STR00212##  
##STR00213## ##STR00214## ##STR00215## ##STR00216## ##STR00217## ##STR00218##  
##STR00219## ##STR00220## ##STR00221## ##STR00222## ##STR00223## ##STR00224##  
##STR00225## ##STR00226## ##STR00227## ##STR00228## ##STR00229## ##STR00230##  
##STR00231## ##STR00232## ##STR00233## ##STR00234## ##STR00235## ##STR00236##  
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(72) ##STR00247## ##STR00248## ##STR00249## ##STR00250## ##STR00251##  
##STR00252## ##STR00253## ##STR00254## ##STR00255## ##STR00256## ##STR00257##  
##STR00258## ##STR00259## ##STR00260## ##STR00261## ##STR00262## ##STR00263##  
##STR00264## ##STR00265## ##STR00266## ##STR00267## ##STR00268## ##STR00269##  
##STR00270## ##STR00271## ##STR00272## ##STR00273## ##STR00274## ##STR00275##  
##STR00276## ##STR00277## ##STR00278## ##STR00279## ##STR00280## ##STR00281##  
##STR00282## ##STR00283## ##STR00284## ##STR00285## ##STR00286## ##STR00287##  
##STR00288## ##STR00289## ##STR00290## ##STR00291## ##STR00292## ##STR00293##  
##STR00294## ##STR00295## ##STR00296## ##STR00297## ##STR00298## ##STR00299##  
##STR00300##  
(73) ##STR00301## ##STR00302## ##STR00303## ##STR00304## ##STR00305##  
##STR00306## ##STR00307## ##STR00308## ##STR00309## ##STR00310## ##STR00311##  
##STR00312## ##STR00313##

(74) In addition, by introducing various substituents to the structure of Chemical Formula 1, compounds having unique properties of the introduced substituents may be synthesized. For example, by introducing substituents normally used as hole injection layer materials, hole transfer layer materials, light emitting layer materials, electron transfer layer materials and charge generation layer materials used for manufacturing an organic light emitting device to the core structure, materials satisfying conditions required for each organic material layer may be synthesized.

(75) In addition, by introducing various substituents to the structure of Chemical Formula 1, the energy band gap may be finely adjusted, and by the high T1 value obtained therefrom, an organic light emitting device with superior efficiency may be provided.

(76) One embodiment of the present specification provides an organic light emitting device including a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes the heterocyclic compound represented by Chemical Formula 1.

(77) In one embodiment of the present specification, the first electrode may be an anode, and the second electrode may be a cathode.

(78) In another embodiment of the present specification, the first electrode may be a cathode, and the second electrode may be an anode.

(79) In one embodiment of the present specification, the organic light emitting device may be a blue organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the blue organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a host material of a light emitting layer of the blue organic light emitting device.

(80) In another embodiment of the present specification, the organic light emitting device may be a green organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the green organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a host material of a light emitting layer of the green organic light emitting device.

(81) In another embodiment of the present specification, the organic light emitting device may be a red organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the red organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a host material of a light emitting layer of the red organic light emitting device.

(82) Specific descriptions on the heterocyclic compound represented by Chemical Formula 1 are the same as the descriptions provided above.

(83) The organic light emitting device of the present specification may be manufactured using common organic light emitting device manufacturing methods and materials except that one or more of the organic material layers are formed using the heterocyclic compound described above.

(84) The heterocyclic compound may be formed into an organic material layer through a solution coating method as well as a vacuum deposition method when manufacturing the organic light emitting device. Herein, the solution coating method means spin coating, dip coating, inkjet printing, screen printing, a spray method, roll coating and the like, but is not limited thereto.

(85) The organic material layer of the organic light emitting device of the present specification may be formed in a single layer structure, but may be formed in a multilayer structure in which two or more organic material layers are laminated. For example, the organic light emitting device of the present disclosure may have a structure including a hole injection layer, a hole transfer layer, a light emitting layer, an electron transfer layer, an electron injection layer and the like as the organic material layer. However, the structure of the organic light emitting device is not limited thereto, and may include a smaller number of organic material layers.

(86) In the organic light emitting device of the present specification, the organic material layer includes an electron transfer layer, and the electron transfer layer may include the heterocyclic compound of Chemical Formula 1. When using the heterocyclic compound as an electron transfer material, HOMO and LUMO may be adjusted depending on the substituent position, and excellent electron transfer efficiency is obtained. In addition, compared to existing electron transfer materials, migration of electrons in a reverse direction is prevented due to a difference in the T1 value, and as a result, superior device efficiency is obtained.

(87) The organic light emitting device of the present disclosure may further include one, two or more layers selected from the group consisting of a light emitting layer, a hole injection layer, a hole transfer layer, an electron injection layer, an electron transfer layer, an electron blocking layer and a hole blocking layer.

(88) FIG. 1 to FIG. 3 illustrate a lamination order of electrodes and organic material layers of an organic light emitting device according to one embodiment of the present specification. However, the scope of the present application is not limited to these diagrams, and structures of organic light emitting devices known in the art may also be used in the present application.

(89) FIG. 1 illustrates an organic light emitting device in which an anode (200), an organic material layer (300) and a cathode (400) are consecutively laminated on a substrate (100). However, the structure is not limited to such a structure, and as illustrated in FIG. 2, an organic light emitting device in which a cathode, an organic material layer and an anode are consecutively laminated on a substrate may also be obtained.

(90) FIG. 3 illustrates a case of the organic material layer being a multilayer. The organic light emitting device according to FIG. 3 includes a hole injection layer (301), a hole transfer layer (302), a light emitting layer (303), an electron transfer layer (304) and an electron injection layer (305). However, the scope of the present application is not limited to such a lamination structure, and as necessary, layers other than the light emitting layer may not be included, and other necessary functional layers may be further included.

(91) The organic material layer including the heterocyclic compound represented by Chemical Formula 1 may further include other materials as necessary.

(92) The organic light emitting device according to one embodiment of the present specification includes a first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and a second electrode provided on the second stack, wherein the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

(93) The organic light emitting device according to one embodiment of the present specification includes a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes two or more stacks, and the two or more stacks each independently include a light emitting layer, a charge generation layer is included between the two or more stacks, and the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

(94) The organic light emitting device according to one embodiment of the present specification includes a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes a first stack including a first light emitting layer; a charge generation layer provided on the first stack; and a second stack including a second light emitting layer, and the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

(95) In addition, the organic light emitting device according to one embodiment of the present specification includes an anode, a cathode, and two or more stacks provided between the anode and the cathode, the two or more stacks each independently include a light emitting layer, a charge generation layer is included between the two or more stacks, and the charge generation layer includes the heterocyclic compound represented by Chemical Formula 1.

(96) In addition, the organic light emitting device according to one embodiment of the present specification includes an anode, a first stack provided on the anode and including a first light emitting layer, a charge generation layer provided on the first stack, a second stack provided on the charge generation layer and including a second light emitting layer, and a cathode provided on the second stack. Herein, the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1. When the heterocyclic compound is included in the charge generation layer, superior efficiency as a charge generation layer material is obtained by an electron-friendly substituent structure and a hole migration-friendly quinoline-dibenzofuran or quinoline-dibenzothiophene fused structure.

(97) The organic light emitting device according to one embodiment of the present specification

includes a first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and a second electrode provided on the second stack, wherein the charge generation layer is an N-type charge generation layer, and the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

(98) In addition, the first stack and the second stack may each independently further include one or more types of the hole injection layer, the hole transfer layer, the hole blocking layer, the electron transfer layer, the electron injection layer and the like described above.

(99) The charge generation layer may be an N-type charge generation layer or a P-type charge generation layer, and the charge generation layer may further include a dopant known in the art in addition to the heterocyclic compound represented by Chemical Formula 1.

(100) As the organic light emitting device according to one embodiment of the present specification, an organic light emitting device having a 2-stack tandem structure is illustrated in FIG. 4.

(101) Herein, the first electron blocking layer, the first hole blocking layer, the second hole blocking layer and the like described in FIG. 4 may not be included in some cases.

(102) In the organic light emitting device according to one embodiment of the present specification, materials other than the heterocyclic compound represented by Chemical Formula 1 are illustrated below, however, these are for illustrative purposes only and not for limiting the scope of the present application, and the materials may be replaced by materials known in the art.

(103) As the anode material, materials having relatively large work function may be used, and transparent conductive oxides, metals, conductive polymers or the like may be used. Specific examples of the anode material include metals such as vanadium, chromium, copper, zinc and gold, or alloys thereof; metal oxides such as zinc oxide, indium oxide, indium tin oxide (ITO) and indium zinc oxide (IZO); combinations of metals and oxides such as ZnO:Al or SnO<sub>2</sub>:Sb; conductive polymers such as poly(3-methylthiophene), poly[3,4-(ethylene-1,2-dioxy)thiophene] (PEDOT), polypyrrole and polyaniline, and the like, but are not limited thereto.

(104) As the cathode material, materials having relatively small work function may be used, and metals, metal oxides, conductive polymers or the like may be used. Specific examples of the cathode material include metals such as magnesium, calcium, sodium, potassium, titanium, indium, yttrium, lithium, gadolinium, aluminum, silver, tin and lead, or alloys thereof; multilayer structure materials such as LiF/Al or LiO<sub>2</sub>/Al, and the like, but are not limited thereto.

(105) As the hole injection material, known hole injection materials may be used, and for example, phthalocyanine compounds such as copper phthalocyanine disclosed in U.S. Pat. No. 4,356,429, or starburst-type amine derivatives such as tris(4-carbazoyl-9-ylphenyl)amine (TCTA), 4,4',4''-tri[phenyl(m-tolyl)amino]triphenylamine (m-MTDATA) or 1,3,5-tris[4-(3-methylphenylphenylamino)phenyl]benzene (m-MTDAPB) described in the literature [Advanced Material, 6, p. 677 (1994)], polyaniline/dodecylbenzene sulfonic acid, poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate), polyaniline/camphor sulfonic acid or polyaniline/poly(4-styrenesulfonate) that are conductive polymers having solubility, and the like, may be used.

(106) As the hole transfer material, pyrazoline derivatives, arylamine-based derivatives, stilbene derivatives, triphenyldiamine derivatives and the like may be used, and low molecular or high molecular materials may also be used.

(107) As the electron transfer material, metal complexes of oxadiazole derivatives, anthraquinodimethane and derivatives thereof, benzoquinone and derivatives thereof, naphthoquinone and derivatives thereof, anthraquinone and derivatives thereof, tetracyanoanthraquinodimethane and derivatives thereof, fluorenone derivatives, diphenyldicyanoethylene and derivatives thereof, diphenoquinone derivatives, 8-hydroxyquinoline

and derivatives thereof, and the like, may be used in addition to the heterocyclic compound, and high molecular materials may also be used as well as low molecular materials.

(108) As examples of the electron injection material, LiF is typically used in the art, however, the present application is not limited thereto.

(109) As the light emitting material, red, green or blue light emitting materials may be used, and as necessary, two or more light emitting materials may be mixed and used. Herein, two or more light emitting materials may be used by being deposited as individual sources of supply or by being premixed and deposited as one source of supply. In addition, fluorescent materials may also be used as the light emitting material, however, phosphorescent materials may also be used. As the light emitting material, materials emitting light by bonding electrons and holes injected from an anode and a cathode, respectively, may be used alone, however, materials having a host material and a dopant material involving in light emission together may also be used.

(110) When mixing light emitting material hosts, same series hosts may be mixed, or different series hosts may be mixed. For example, any two or more types of materials among n-type host materials or p-type host materials may be selected and used as a host material of a light emitting layer.

(111) The organic light emitting device according to one embodiment of the present specification may be a top-emission type, a bottom-emission type or a dual-emission type depending on the materials used.

(112) The heterocyclic compound according to one embodiment of the present specification may also be used in an organic electronic device including an organic solar cell, an organic photo conductor, an organic transistor and the like under a similar principle used in the organic light emitting device.

(113) Hereinafter, the present specification will be described in more detail with reference to examples, however, these are for illustrative purposes only, and the scope of the present application is not limited thereto.

[Preparation Example 1] Preparation of Compound 4

(114) ##STR00314## ##STR00315##

1) Preparation of Compound 1-1

(115) After dissolving 1-bromo-8-chlorodibenzo[b,d]furan (100 g, 0.3552 mol) and 2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline (77.8 g, 0.3552 mol) in toluene, ethanol and H.sub.2O (2000 mL:200 mL:200 mL), Pd(PPh.sub.3).sub.4 (tetrakis(triphenylphosphine)palladium(0)) (20.5 g, 0.017 mol) and NaHCO.sub.3 (89 g, 1.06 mol) were introduced thereto, and the result was stirred for 3 hours at 100° C. After the reaction was completed, methylene chloride (MC) and distilled water were introduced to the reaction solution, and the result was extracted, then dried with anhydrous MgSO.sub.4, and the solvent was removed using a rotary evaporator to obtain Compound 1-1 (77 g, 74%) in a liquid form.

(116) 2) Preparation of Compound 1-2

(117) Compound 1-1 (77 g, 0.262 mol) and triethylamine (86 mL, 0.393 mol) were introduced to methylene chloride (1200 mL) and dissolved therein. To the mixture, benzoyl chloride (55 g, 0.393 mol) dissolved in methylene chloride (300 mL) was slowly added dropwise at 0° C. After the reaction was completed, white solids in the reaction solution were filtered and washed with hexane. These were dried to obtain Compound 1-2 (86 g, 83%) in a solid form.

(118) 3) Preparation of Compound 1-3

(119) After dissolving Compound 1-2 (86 g, 0.216 mol) in nitrobenzene (1500 mL), POCl.sub.3 (30 mL, 0.324 mol) was slowly added dropwise thereto, and the result was reacted for 15 hours at 140° C. After the reaction was completed, a solution obtained by dissolving NaHCO.sub.3 in distilled water was slowly introduced to the reaction solution, and the result was stirred. Solids produced after that were filtered and collected, and the collected solids were recrystallized in MC and MeOH to obtain Compound 1-3 (41 g, 51%) in a solid form.

(120) 4) Preparation of Compound 1-4

(121) After dissolving Compound 1-3 (41 g, 0.1079 mol) in 1,4-dioxane (800 mL), Pd.sub.2(dba).sub.3 (tris(dibenzylideneacetone)dipalladium(0)) (9.8 g, 0.0107 mol), XPhos (2-dicyclohexylphosphino-2',4',6'-triisopropylbiphenyl) (5.1 g, 0.0107 mol) and KOAc (potassium acetate) (31 g, 0.3237 mol) were dissolved therein, and the result was reacted for 4 hours at 90° C. After the reaction was completed, water and methylene chloride were introduced to the reaction solution, and the result was extracted. Solids obtained after that were dried, and recrystallized with methylene chloride and methanol to obtain Compound 1-4 (30 g, 60%) in a solid form.

(122) 5) Preparation of Compound 4

(123) After dissolving Compound 1-4 (6.4 g, 0.017 mol) and 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine (7.4 g, 0.017 mol) in 1,4-dioxane:H.sub.2O=100 mL:20 mL, Pd(PPh.sub.3).sub.4 (1.96 g, 0.0017 mol) and K.sub.3PO.sub.4 (20.8 g, 0.051 mol) were introduced thereto, and the result was stirred for 5 hours at 100° C. After the reaction was completed, solids produced in the reaction solution were washed with 1,4-dioxane and H.sub.2O. After that, only the solids were purified using a recrystallization method in dichlorobenzene (DCB) to obtain Compound 4 (6.6 g, 60%).

[Preparation Example 2] Preparation of Compound 88

(124) ##STR00316## ##STR00317##

1) Preparation of Compound 1-1

(125) Compound 1-1 was the same as Compound 1-1 of Preparation Example 1.

(126) 2) Preparation of Compound 2-2

(127) Compound 1-1 (77 g, 0.262 mol) and triethylamine (86 mL, 0.393 mol) were introduced to MC (1200 mL) and dissolved therein. To the mixture, 1-pyrenecarbonyl chloride (104 g, 0.393 mol) dissolved in methylene chloride (300 mL) was slowly added dropwise at 0° C. After the reaction was completed, white solids in the reaction solution were filtered and washed with hexane. These were dried to obtain Compound 2-2 (75 g, 67%) in a solid form.

(128) 3) Preparation of Compound 2-3

(129) After dissolving Compound 2-2 (86 g, 0.143 mol) in nitrobenzene (1500 mL), POCl.sub.3 (20 mL, 0.215 mol) was slowly added dropwise thereto, and the result was reacted for 15 hours at 140° C. After the reaction was completed, a solution obtained by dissolving NaHCO.sub.3 in distilled water was slowly introduced to the reaction solution, and the result was stirred. Solids produced after that were filtered and collected, and the collected solids were recrystallized in methylene chloride and methanol to obtain Compound 2-3 (29 g, 41%) in a solid form.

(130) 4) Preparation of Compound 2-4

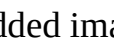
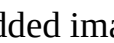
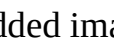
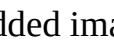
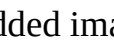
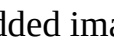
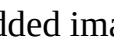
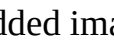
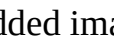
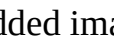
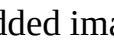
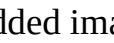
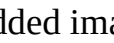
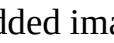
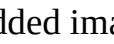
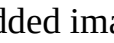
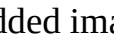
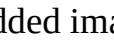
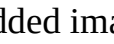
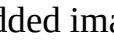
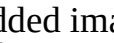
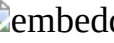
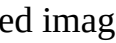
(131) After dissolving Compound 2-3 (29 g, 0.057 mol) in 1,4-dioxane (800 mL), Pd.sub.2(dba).sub.3 (5.2 g, 0.0057 mol), XPhos (2.7 g, 0.0057 mol) and KOAc (16.7 g, 0.171 mol) were dissolved therein, and the result was reacted for 4 hours at 90° C. After the reaction was completed, water and MC were introduced to the reaction solution, and the result was extracted. After that, solids obtained by drying the result were recrystallized with methylene chloride and methanol to obtain Compound 2-4 (20 g, 60%) in a solid form.

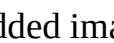
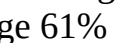
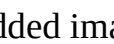
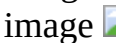
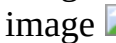
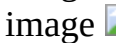
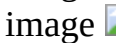
(132) 5) Preparation of Compound 88

(133) After dissolving Compound 2-4 (10 g, 0.017 mol) and 4-(3-bromophenyl)dibenzo[b,d]thiophene (5.7 g, 0.017 mol) in 1,4-dioxane:H.sub.2O=100 mL:20 mL, Pd(PPh.sub.3).sub.4 (1.96 g, 0.0017 mol) and K.sub.3PO.sub.4 (20.8 g, 0.051 mol) were introduced thereto, and the result was stirred for 5 hours at 100° C. After the reaction was completed, solids produced in the reaction solution were washed with 1,4-dioxane and H.sub.2O. After that, only the solids were purified using a recrystallization method in dichlorobenzene (DCB) to obtain Compound 88 (4.9 g, 40%).

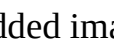
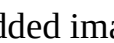
(134) Target compounds were synthesized in the same manner as in Preparation Example 1 except that Intermediate A of the following Table 1 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-

1,3,5-triazine.

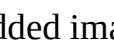
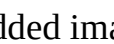
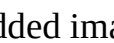
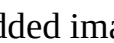
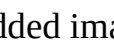
| (135) TABLE-US-00001 TABLE 1 Compound Intermediate A Target Compound Yield       |                                                                                    |     |      | 1                                                                                  |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----|------|------------------------------------------------------------------------------------|
|   |    | 61% | 50   |    |
|  |   | 51% | 8    |   |
|  |   | 61% | 11   |   |
|  |   | 51% | 13   |   |
|  |   | 61% | 16   |   |
|  |   | 51% | 180  |   |
|  |   | 61% | 25   |   |
|  |   | 51% | 30   |   |
|  |   | 61% | 36   |   |
|  |   | 51% | 38   |   |
|  |   | 61% | 420  |   |
|  |   | 51% | 45   |   |
|  |   | 61% | 52   |   |
|  |   | 51% | 56   |   |
|  |   | 61% | 64   |   |
|  |   | 51% | 690  |   |
|  |   | 61% | 72   |   |
|  |   | 51% | 77   |   |
|  |   | 61% | 80   |   |
|  |   | 51% | 86   |   |
|  |   | 62% | 9100 |   |
| 9202                                                                             |  | 60% | 9704 |  |

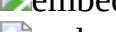
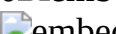
|                                                                                    |                                                                                     |       |                                                                                     |                                                                                     |     |       |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|-------|
|   | 61%                                                                                 | 10006 |   |   | 62% | 10208 |
|  | 51%                                                                                 | 1030  |  |  | 60% |       |
| 107                                                                                |  | 61%   | 112                                                                                 |  | 62% |       |
| 113                                                                                |  | 51%   | 114                                                                                 |  | 60% |       |
| 1160                                                                               |  | 61%   | 119                                                                                 |  | 62% |       |
| 120                                                                                |  | 51%   | 132                                                                                 |  | 62% |       |
| 133                                                                                |  | 51%   | 1350                                                                                |  | 60% |       |
| 139                                                                                |  | 61%   | 140                                                                                 |  | 62% |       |
| 141                                                                                |  | 51%   | 142                                                                                 |  | 60% |       |

(136) Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 1 of the following Table 2 was used instead of 1-pyrenecarbonyl chloride, and Intermediate B of the following Table 2 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

| (137) TABLE-US-00002 TABLE 2 Intermediate B Intermediate 1 Target Compound         |                                                                                     |                                                                                     |     | 1210 |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|------|
|  |  |  | 60% | 128  |
|  |  | 61%                                                                                 |     |      |

(138) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 6-bromo-2-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate C of the following Table 3 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

| (139) TABLE-US-00003 TABLE 3 Compound Intermediate C Target Compound Yield         |                                                                                     |     |      | 58                                                                                  |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|------|-------------------------------------------------------------------------------------|
|  |  | 62% | 156  |  |
|  |  | 65% | 159  |  |
|  |  | 51% | 165  |  |
|  |  | 61% | 1700 |  |
|  |  | 65% | 172  |  |

 embedded image 62% 178  embedded image 60% 179 0  
 embedded image 61% 180  embedded image 60% 185  
 embedded image 62% 187  embedded image 65% 190  
 embedded image 65% 191 0  embedded image 62% 192  
 embedded image 51% 194  embedded image 60% 195  
 embedded image 61% 196  embedded image 60% 200 0  
 embedded image 55%

(140) Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 2 of the following Table 4 was used instead of 1-pyrenecarbonyl chloride, 6-bromo-2-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate D of the following Table 4 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

(141) TABLE-US-00004 TABLE 4 Intermediate D Intermediate 2 Target Compound 168  
 embedded image  embedded image  embedded image 60% 176  embedded image  
 embedded image  embedded image 61%

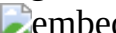
(142) A target compound was synthesized in the same manner as in Preparation Example 1 except that 2-bromo-8-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate E of the following Table 5 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

(143) TABLE-US-00005 TABLE 5 Compound Intermediate E Target Compound Yield 149  
 embedded image  embedded image 52%

(144) A target compound was synthesized in the same manner as in Preparation Example 2 except that Intermediate 3 of the following Table 6 was used instead of 1-pyrenecarbonyl chloride, 2-bromo-8-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate F of the following Table 6 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

(145) TABLE-US-00006 TABLE 6 Compound Intermediate F Intermediate 3 Target Compound Yield 145 00  embedded image 01  embedded image 02  embedded image 53%

(146) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 1-bromo-6-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate G of the following Table 7 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

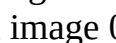
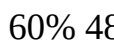
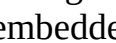
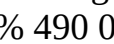
(147) TABLE-US-00007 TABLE 7 Compound Intermediate G Target Compound Yield 384 03  
 embedded image 04  embedded image 53% 386 05  embedded image 06  embedded image 62% 387 07  embedded image 08  embedded image 65% 388 09  embedded image 0  
 embedded image 65% 392  embedded image  embedded image 62% 395  embedded image  
 embedded image 51% 396  embedded image  embedded image 60% 397  embedded image  
 embedded image 61% 399  embedded image 0  embedded image 60% 400  embedded image  
 embedded image 62% 401  embedded image  embedded image 65% 404  embedded image  
 embedded image 65% 408  embedded image  embedded image 51% 409  embedded image 0  
 embedded image 60% 411  embedded image  embedded image 61% 413  embedded image  
 embedded image 60% 415  embedded image  embedded image 62% 416  embedded image  
 embedded image 65% 417  embedded image 0  embedded image 65%

(148) A target compound was synthesized in the same manner as in Preparation Example 2 except that Intermediate 4 of the following Table 8 was used instead of 1-pyrenecarbonyl chloride, 1-bromo-6-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate H of the following Table 8 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

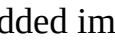
(149) TABLE-US-00008 TABLE 8 Compound Intermediate H Intermediate 4 Target Compound Yield 407  embedded image  embedded image  embedded image 53%

(150) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 4,6-dibromodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate I of the following Table 9 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

(151) TABLE-US-00009 TABLE 9 Compound Intermediate I Target Compound Yield 421

|                                                                                    |                                                                                     |         |                                                                                     |                                                                                     |         |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------|
|    |    | 53% 423 |    |    | 62% 424 |
|    |    | 65% 426 |    |    | 65% 431 |
|    |    | 62% 432 |    |    | 51% 433 |
|    |    | 60% 436 |    |    | 61% 438 |
|    |    | 60% 440 |    |    | 62% 441 |
|    |    | 65% 443 |    |    | 65% 447 |
|    |    | 62% 448 |    |    | 51% 449 |
|    |    | 60% 450 |    |    | 61% 452 |
|    |    | 60% 456 |    |    | 62% 460 |
|    |    | 65% 462 |    |    | 65% 465 |
|    |    | 62% 467 |    |    | 51% 470 |
|    |    | 61% 475 |    |    | 60% 476 |
|    |    | 62% 477 |    |    | 65% 479 |
|    |    | 65% 481 |    |    | 62% 482 |
|    |    | 51% 485 |    |    | 60% 476 |
|    |    | 61% 489 |    |    | 62% 482 |
|    |    | 62% 493 |    |    | 65% 495 |
|    |    | 65% 494 |    |    | 65% 495 |
|    |    | 62% 500 |    |    | 51% 504 |
|   |   | 60% 505 |   |   | 61% 506 |
|  |  | 60% 507 |  |  | 62% 512 |
|  |  | 65% 513 |  |  | 65% 516 |
|  |  | 62% 517 |  |  | 51% 518 |
|  |  | 60% 519 |  |  | 61% 520 |
|  |  | 60% 524 |  |  | 65% 525 |
|  |  | 65% 527 |  |  | 62% 528 |
|  |  | 51% 531 |  |  | 60% 532 |
|  |  | 61% 533 |  |  | 60% 535 |
|  |  | 62% 536 |  |  | 65% 539 |
|  |  | 65% 542 |  |  | 62% 543 |
|  |  | 51% 544 |  |  | 60% 547 |
|  |  | 61% 549 |  |  | 62% 550 |
|  |  | 65%     |                                                                                     |                                                                                     |         |

(152) Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 5 of the following Table 10 was used instead of 1-pyrenecarbonyl chloride, 4,6-dibromodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate I of the following Table 10 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

(153) TABLE-US-00010 TABLE 10 Compound Intermediate I Target Compound Yield 468    53% 522   
  52% 548   0  
 50%

(154) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 1-chloro-6-fluorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate J of the following Table 11 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-

1,3,5-triazine.

(155) TABLE-US-00011 TABLE 11 Compound Intermediate J Target Compound Yield 554

53% 555 62% 557  
 65% 558 65% 564  
 62% 565 51% 566  
 60% 569 61% 573  
 60% 574 62%  
580 01 65% 583 03 04  
 65% 584 05 62% 585 07

51% 587 09 60%  
588 61% 594 60%  
(156) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 7-chloro-1-fluorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate K of the following Table 12 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

(157) TABLE-US-00012 TABLE 12 Compound Intermediate K Target Compound Yield 597

53% 598 62% 599  
 65% 600 65% 601  
 62% 605 51% 606  
 60% 611 61% 612  
 60% 615 51% 618  
 50%

(158) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 1-bromo-8-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate L of the following Table 13 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

(159) TABLE-US-00013 TABLE 13 Compound Intermediate L Target Compound Yield 623

62% 624 65% 626  
 65% 629 62% 632  
 51% 634 60% 635  
 61% 636 60% 640  
 62% 641 65% 644  
 65% 646 62% 649  
 51% 651 60% 652  
 61% 653 60% 654  
 62% 655 65% 657  
 65% 659 62% 660  
 51% 661 60% 664  
 61% 665 60% 666  
 62% 670 65% 674  
 62% 675 51% 677  
 60% 680 61% 682  
 60% 685 65%  
687 01 65% 688 03 04  
 62% 691 05 51% 695 07  
 61% 696 09 62%  
697 65% 700 65%  
704 51% 707 60%  
708 62% 709 65%  
713 65% 714 62%

716  51% 717  60%  
721  61% 728  60%  
729  62% 731  65%  
732  65% 733  62%  
734  51% 735  60%  
736  61% 741  60%  
743  62% 745  65%  
748  65% 752  51%  
756  60% 757  61%  
761  60%

(160) Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 6 of the following Table 14 was used instead of 1-pyrenecarbonyl chloride, 1-bromo-8-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate M of the following Table 14 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

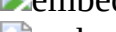
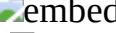
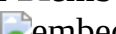
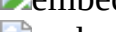
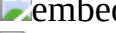
(161) TABLE-US-00014 TABLE 14 Compound Intermediate M Intermediate 6 Target Compound Yield 620   53% 672   
 0  62% 683    
 60% 692   60% 703  
  62% 750 0   
  52%

(162) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 2-bromo-8-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate N of the following Table 15 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

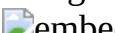
(163) TABLE-US-00015 TABLE 15 Compound Intermediate N Target Compound Yield 764  
  53% 766   62% 767  
  65% 769  0  65%

(164) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 6-bromo-2-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate O of the following Table 16 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

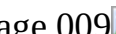
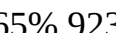
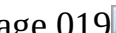
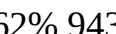
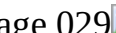
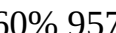
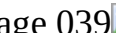
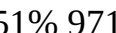
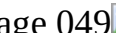
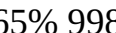
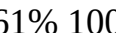
(165) TABLE-US-00016 TABLE 16 Compound Intermediate O Target Compound Yield 771  
  62% 773   62% 775  
  65% 783   62% 784  
 00  51% 787 01  02   
61% 789 03  04  60% 791 05  06  
 62% 792 07  08  65% 793 09  
 0  65% 798   62% 801  
  51% 802   60% 807  
  61% 809  0  60% 810  
  62% 814   65% 816  
  65% 817   62% 818  
 0  51% 822   60% 823  
  61% 829   62% 830  
  65% 831  0  65% 833  
  62% 836   51% 837  
  60% 838   61% 842  
 0  60% 844   62% 847  
  65% 848   65% 849

 embedded image 62% 858  embedded image 0  embedded image 51% 860  
 embedded image 60% 861  embedded image 61% 862  
 embedded image 60% 863  embedded image 62% 870  
 embedded image 0  embedded image 65% 871  embedded image 65% 872  
 embedded image 62% 883  embedded image 51% 884  
 embedded image 60% 887  embedded image 0  embedded image 61% 888  
 embedded image 60% 890  embedded image 62% 893  
 embedded image 65% 905  embedded image 65% 908  
 embedded image 0  embedded image 62% 910  embedded image 51% 911  
 embedded image 60%

(166) Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 7 of the following Table 17 was used instead of 1-pyrenecarbonyl chloride, 6-bromo-2-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate P of the following Table 17 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

(167) TABLE-US-00017 TABLE 17 Compound Intermediate P Intermediate 7 Target Compound Yield 779  embedded image  embedded image  embedded image 56% 786  embedded image  embedded image 000  embedded image 56% 825 001  embedded image 002  embedded image 003  embedded image 60%

(168) Target compounds were synthesized in the same manner as in Preparation Example 1 except that 1-bromo-6-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate Q of the following Table 18 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

(169) TABLE-US-00018 TABLE 18 Compound Intermediate Q Target Compound Yield 916 004  embedded image 005  embedded image 53% 918 006  embedded image 007  embedded image 62% 919 008  embedded image 009  embedded image 65% 920 010  embedded image 011  embedded image 65% 923 012  embedded image 013  embedded image 62% 924 014  embedded image 015  embedded image 51% 930 016  embedded image 017  embedded image 60% 931 018  embedded image 019  embedded image 61% 932 020  embedded image 021  embedded image 62% 943 022  embedded image 023  embedded image 65% 945 024  embedded image 025  embedded image 65% 952 026  embedded image 027  embedded image 62% 955 028  embedded image 029  embedded image 51% 956 030  embedded image 031  embedded image 60% 957 032  embedded image 033  embedded image 62% 960 034  embedded image 035  embedded image 65% 963 036  embedded image 037  embedded image 65% 965 038  embedded image 039  embedded image 62% 968 040  embedded image 041  embedded image 51% 971 042  embedded image 043  embedded image 62% 979 044  embedded image 045  embedded image 65% 980 046  embedded image 047  embedded image 65% 984 048  embedded image 049  embedded image 62% 986 050  embedded image 051  embedded image 65% 998 052  embedded image 053  embedded image 65% 991 054  embedded image 055  embedded image 62% 995 056  embedded image 057  embedded image 51% 1000 058  embedded image 059  embedded image 60% 1004 060  embedded image 061  embedded image 61% 1006 062  embedded image 063  embedded image 60% 1011 064  embedded image 065  embedded image 62% 1015 066  embedded image 067  embedded image 65%

(170) Synthesis identification results for the compounds prepared as above are shown in the following Table 19.

(171) TABLE-US-00019 TABLE 19 Compound FD-MS Compound FD-MS 1 m/z = 576.65 4 m/z = 652.75 (C40H24N4O = 576.195) (C46H28N4O = 652.226) 5 m/z = 728.85 7 m/z = 804.95 (C52H32N4O = 728.25) (C58H36N4O = 804.28) 8 m/z = 728.85 9 m/z = 804/95 (C52H32N4O = 728.25) (C58H36N4O = 804.28) 11 m/z = 666.74 12 m/z = 682.80 (C46H26N4O2 = 666.20)

(C46H26N4OS = 682.18) 13 m/z = 741.85 15 m/z = 665.75 (C52H31N5O = 741.25)  
(C46H27N5O = 665.22) 16 m/z = 742.83 17 m/z = 792.80 (C52H30N4O = 742.23) (C56H32N4O2  
= 792.25) 18 m/z = 842.95 24 m/z = 571.67 (C60H34N4O2 = 842.26) (C43H25NO = 571.19) 25  
m/z = 521.61 28 m/z = 577.70 (C39H23NO = 521.17) (C41H23NOS = 577.15) 30 m/z = 628.77 35  
m/z = 818.93 (C46H32N2O = 628.25) (C58H34N4O2 = 818.26) 36 m/z = 818.93 37 m/z = 752.92  
(C58H34N4O2 = 818.26) (C56H36N2O = 752.28) 38 m/z = 599.69 41 m/z = 725.85  
(C43H25N3O = 599.2) (C53H1N3O = 725.24) 42 m/z = 652.75 44 m/z = 659.78 (C46H28N4O =  
652.22) (C50H29NO = 659.22) 45 m/z = 621.67 47 m/z = 549.63 (C43H28NO2P = 621.18)  
(C39H23N3O = 549.18) 52 m/z = 618.75 53 m/z = 660.77 (C43H26N2OS = 618.17)  
(C49H28N2O = 660.22) 56 m/z = 751.88 57 m/z = 834.99 (C55H33N3O = 751.26)  
(C58H34N4OS = 834.24) 58 m/z = 758.89 64 m/z = 804.95 (C52H30N4OS = 758.21)  
(C58H36N4O = 804.28) 68 m/z = 732.86 69 m/z = 732.86 (C50H28N4OS = 732.19)  
(C50H28N4OS = 732.19) 70 m/z = 868.01 72 m/z = 690.76 (C62H37N5O = 867.30)  
(C47H26N6O = 690.21) 75 m/z = 742.83 77 m/z = 758.89 (C52H30N4O2 = 742.23)  
(C52H30N4OS = 758.21) 79 m/z = 768.92 80 m/z = 741.85 (C55H36N4O = 768.28) (C5231N5O  
= 741.25) 84 m/z = 660.77 86 m/z = 597.71 (C49H28N2O = 660.22) (C45H27NO = 597.20) 88  
m/z = 727.88 89 m/z = 663.77 (C53H29NOS = 727.19) (C49H29NO2 = 663.22) 91 m/z = 752.91  
92 m/z = 675.79 (C56H36N2O = 752.28) (C49H29N3O = 675.23) 97 m/z = 575.67 100 m/z =  
727.86 (C41H25N3O = 575.20) (C53H33N3O = 727.26) 102 m/z = 701.82 103 m/z = 727.86  
(C51H31N3O = 701.24) (C53H33N3O = 727.26) 107 m/z = 727.86 112 m/z = 681.81  
(C53H33N3O = 727.26) (C47H27N3OS = 681.18) 113 m/z = 740.86 114 m/z = 691.83  
(C53H32N4O = 740.25) (C50H33N3O = 691.26) 116 m/z = 741.85 119 m/z = 817.99  
(C53H31N3O2 = 741.24) (C60H39N3O = 817.30) 120 m/z = 740.86 121 m/z = 725.85  
(C53H32N4O = 740.25) (C53H31N3O = 725.24) 128 m/z = 777.92 132 m/z = 681.81  
(C57H35N3O = 777.27) (C47H27N3OS = 681.18) 133 m/z = 681.81 135 m/z = 664.76  
(C47H27N3OS = 681.18) (C47H28N4O = 664.22) 139 m/z = 757.91 140 m/z = 816.96  
(C53H31N3OS = 757.21) (C59H36N4O = 816.28) 141 m/z = 767.93 142 m/z = 740.86  
(C56H37N3O = 767.29) (C53H32N4O = 740.25) 145 m/z = 702.81 149 m/z = 728.85  
(C50H30N4O = 702.24) (C52H32N4O = 728.25) 156 m/z = 652.75 157 m/z = 652.75  
(C46H26N4O = 652.22) (C46H28N4O = 652.22) 159 m/z = 804.95 161 m/z = 666.74  
(C58H36N4O = 804.28) (C46H26N4O2 = 666.20) 165 m/z = 682.80 166 m/z = 741.85  
(C46H26N4OS = 682.18) (C52H31N5O = 741.25) 168 m/z = 715.81 170 m/z = 842.95  
(C50H29N4O = 715.23) (C60H34N4O2 = 842.26) 171 m/z = 758.89 172 m/z = 817.95  
(C52H30N4OS = 758.21) (C58H35N5O = 817.28) 173 m/z = 768.92 176 m/z = 571.67  
(C55H36N4O = 768.28) (C43H25NO = 571.19) 178 m/z = 573.69 179 m/z = 527.64 (C43H27NO  
= 573.20) (C37H21NOS = 527.13) 180 m/z = 552.67 185 m/z = 895.03 (C40H28N2O = 552.22)  
(C64H38N4O2 = 894.29) 187 m/z = 676.81 190 m/z = 675.79 (C50H32N2O = 676.25)  
(C49H29N3O = 675.23) 191 m/z = 725.85 192 m/z = 652.75 (C53H31N3O = 725.24)  
(C46H28N4O = 652.22) 194 m/z = 659.78 195 m/z = 548.64 (C50H29NO = 659.22) (C40H24N2O  
= 548.18) 196 m/z = 625.73 200 m/z = 659.78 (C45H27N3O = 625.21) (C50H29NO = 659.22) 202  
m/z = 660.77 204 m/z = 802.93 (C49H28N2O = 660.22) (C58H34N4O = 802.27) 207 m/z =  
728.85 210 m/z = 742.83 (C52H32N4O = 728.25) (C52H30N4O2 = 742.23) 213 m/z = 732.86 214  
m/z = 868.01 (C50H28N4OS = 732.19) (C62H37N5O = 867.30) 216 m/z = 717.83 217 m/z =  
690.76 (C50H31N5O = 717.25) (C47H26N6O = 690.21) 220 m/z = 742.83 222 m/z = 758.89  
(C52H30N4O2 = 742.23) (C52H30N4OS = 758.21) 223 m/z = 817.95 224 m/z = 768.92  
(C58H35N5O = 817.28) (C55H36N4O = 768.28) 225 m/z = 741.85 227 m/z = 602.69  
(C52H31N5O = 741.25) (C43H26N2O2 = 602.19) 228 m/z = 618.75 229 m/z = 660.77  
(C43H26N2OS = 618.17) (C49H26N2O = 660.22) 233 m/z = 911.09 237 m/z = 675.79  
(C64H38N4OS = 910.27) (C49H29N3O = 675.23) 240 m/z = 662.79 243 m/z = 651.76  
(C49H30N2O = 662.23) (C47H29N3O = 651.23) 245 m/z = 651.76 247 m/z = 651.76

(C47H29N3O = 651.23) (C47H29N3O = 651.23) 249 m/z = 803.96 251 m/z = 715.81  
(C59H37N3O = 803.29) (C51H29N3O2 = 715.22) 256 m/z = 740.86 257 m/z = 691.83  
(C53H32N4O = 740.25) (C50H33N3O = 691.26) 259 m/z = 791.91 260 m/z = 741.85  
(C57H33N3O2 = 791.25) (C53H31N3O2 = 741.24) 262 m/z = 757.91 265 m/z = 767.93  
(C53H31N3OS = 757.21) (C56H37N3O = 767.29) 266 m/z = 740.86 267 m/z = 575.57  
(C53H32N4O = 740.25) (C41H25N3O = 575.20) 269 m/z = 727.86 271 m/z = 727.86  
(C53H33N3O = 727.26) (C53H33N3O = 727.26) 272 m/z = 803.96 273 m/z = 727.86  
(C59H37N3O = 803.29) (C53H33N3O = 727.26) 274 m/z = 841.97 276 m/z = 665.75  
(C61H35N3O2 = 841.27) (C47H27N3O2 = 665.21) 277 m/z = 681.81 279 m/z = 740.86  
(C47H27N3OS = 681.18) (C53H32N4O = 740.25) 281 m/z = 664.76 284 m/z = 741.85  
(C47H28N4O = 662.22) (C53H31N3O2 = 741.24) 285 m/z = 757.91 287 m/z = 816.96  
(C53H31N3OS = 752.21) (C59H36N4O = 816.28) 288 m/z = 767.93 292 m/z = 728.85  
(C56H37N3O = 767.29) (C52H32N4O = 728.25) 294 m/z = 855.01 296 m/z = 652.75  
(C62H38N4O = 854.30) (C46H28N4O = 652.22) 299 m/z = 682.80 302 m/z = 665.75  
(C46H26N4OS = 682.18) (C46H27CN5O = 665.22) 304 m/z = 742.83 305 m/z = 758/89  
(C52H30N4O2 = 742.23) (C52H30N4OS = 758.21) 307 m/z = 817.95 308 m/z = 768.92  
(C58H35N4O = 817.28) (C55H36N4O = 768.28) 309 m/z = 741.85 310 m/z = 571.67  
(C52H31N5O = 741.25) (C43H25NO = 571.19) 312 m/z = 597.71 313 m/z = 753.69 (C45H27NO  
= 597.20) (C43H27NO = 573.20) 315 m/z = 521.61 316 m/z = 597.71 (C39H23NO = 521.17)  
(C45H27NO = 597.20) 318 m/z = 603.73 319 m/z = 742.83 (C43H25NOS = 603.16)  
(C52H30N4O2 = 742.23) 322 m/z = 752.91 325 m/z = 675.79 (C56H36N2O = 752.28)  
(C49H29N3O = 675.23) 326 m/z = 725.85 327 m/z = 652.75 (C53H31N3O = 725.24)  
(C46H28N4O = 652.22) 329 m/z = 548.64 331 m/z = 636.75 (C40H24N2O = 548.18)  
(C47H28N2O = 636.22) 332 m/z = 662.79 336 m/z = 752.87 (C49H30N2O = 662.23)  
(C54H32N4O = 752.25) 337 m/z = 751.88 338 m/z = 693.86 (C55H33N3O = 751.26)  
(C49H31N3S = 693.22) 339 m/z = 691.85 342 m/z = 728.85 (C49H29N3S = 691.20)  
(C52H32N4O = 728.25) 343 m/z = 778.91 345 m/z = 868.01 (C56H34N4O = 778.27)  
(C62H37N5O = 867.30) 348 m/z = 782.92 350 m/z = 717.83 (C54H30N4OS = 782.21)  
(C50H31N5O = 717.25) 353 m/z = 742.83 355 m/z = 758.89 (C52H30N4O2 = 742.23)  
(C52H30N4OS = 758.21) 358 m/z = 768.92 359 m/z = 741.85 (C55H36N4O = 768.28)  
(C52H31N4O = 741.25) 360 m/z = 628.77 364 m/z = 597.71 (C46H32N2O = 628.25) (C45H27NO  
= 597.20) 365 m/z = 652.75 367 m/z = 945.09 (C46H28N4O = 652.22) (C68H40N4O2 = 944.31)  
368 m/z = 713.83 373 m/z = 712.85 (C53H31NO2 = 713.23) (C53H32N2O = 712.25) 374 m/z =  
735.88 377 m/z = 727.86 (C56H33NO = 735.25) (C53H33N3O = 727.26) 378 m/z = 803.95 379  
m/z = 751.88 (C59H27N3O = 803.29) (C55H33N3O = 751.26) 380 m/z = 727.86 384 m/z =  
665.75 (C53H33N3O = 727.26) (C47H27N3O2 = 665.21) 386 m/z = 731.87 387 m/z = 740.86  
(C51H29N3OS = 731.20) (C53H32N4O = 740/25) 388 m/z = 691.83 392 m/z = 741.85  
(C50H33N3O = 691.26) (C53H31N3O2 = 741.24) 395 m/z = 917.08 396 m/z = 740.86  
(C67H40N4O = 916.32) (C53H32N4O = 740.25) 397 m/z = 651.76 399 m/z = 803.96 (C47H2N3O  
= 651.23) (C59H37N3O = 803.29) 400 m/z = 651.76 401 m/z = 727.86 (C47H29N3O = 651.23)  
(C53H33N3O = 727.26) 404 m/z = 665.75 407 m/z = 805.95 (C47H27N3O2 = 665.21)  
(C57H31N3OS = 805.21) 408 m/z = 740.86 409 m/z = 691.83 (C53H32N4O = 740.25)  
(C50H33N3O = 691.25) 411 m/z = 741.85 413 m/z = 741.85 (C53H31N3O2 = 741.24)  
(C53H31N3O2 = 741.24) 415 m/z = 757.91 416 m/z = 816.96 (C53H31N3OS = 757.21)  
(C59H36N4O = 816.28) 417 m/z = 767.93 421 m/z = 804.95 (C56H37N3O = 767.29)  
(C58H36N4O = 804.28) 423 m/z = 702.81 424 m/z = 652.75 (C50H30N4O = 702.24)  
(C46H26N4O = 652.22) 426 m/z = 778.91 431 m/z = 682.80 (C56H34N4O = 778.27)  
(C46H26N4OS = 682.18) 432 m/z = 741.85 433 m/z = 692.82 (C52H31N5O = 741.25)  
(C49H32N4O = 692.25) 436 m/z = 742.83 438 m/z = 758.89 (C52H30N4O2 = 742.23)  
(C52H30N4OS = 758.21) 440 m/z = 741.85 441 m/z = 571.67 (C52H31N5O = 741.25)

(C43H25NO = 571.19) 443 m/z = 597.71 447 m/z = 628.77 (C54H27NO = 597.20) (C46H32N2O = 628.25) 448 m/z = 597.71 449 m/z = 649.79 (C45H27NO = 597.20) (C49H31NO = 649.24) 450 m/z = 911.09 452 m/z = 665.75 (C64H38N4OS = 910.27) (C47H27N3O2 = 665.21) 456 m/z = 675.79 460 m/z = 659.78 (C49H29N3O = 675.23) (C50H29NO = 659.22) 462 m/z = 548.64 465 m/z = 636.75 (C40H24N2O = 548.18) (C47H28N2O = 636.22) 467 m/z = 659.78 468 m/z = 652.75 (C50H29NO = 659.22) (C47H28N2O = 652.21) 470 m/z = 752.87 475 m/z = 728.85 (C54H32N4O = 752.25) (C52H32N4O = 728.25) 476 m/z = 742.83 477 m/z = 742.83 (C52H30N4O2 = 742.23) (C52H30N4O2 = 742.23) 479 m/z = 732.86 481 m/z = 868.01 (C50H28N4OS = 732.19) (C62H37N5O = 867.30) 482 m/z = 717.83 485 m/z = 742.83 (C50H31N5O = 717.25) (C52H30N4O2 = 742.23) 488 m/z = 758.89 489 m/z = 817.95 (C52H30N4OS = 758.21) (C58H35N5O = 817.28) 490 m/z = 818.98 493 m/z = 602.69 (C59H38N4O = 818.30) (C43H26N2O2 = 602.19) 494 m/z = 618.75 495 m/z = 660.77 (C43H26N2OS = 618.17) (C49H28N2O = 660.22) 500 m/z = 663.77 504 m/z = 697.77 (C49H29NO2 = 663.22) (C49H32NO2P = 697.21) 505 m/z = 762.91 506 m/z = 735.88 (C57H34N2O = 762.26) (C56H33NO = 735.25) 509 m/z = 727.86 512 m/z = 727.86 (C53H33N3O = 727.26) (C53H33N3O = 727.26) 513 m/z = 665.75 516 m/z = 681.81 (C47H27N3O2 = 556.21) (C47H27N3OS = 681.18) 517 m/z = 681.81 518 m/z = 740.86 (C47H27N3OS = 681.18) (C53H32N4O = 740.25) 519 m/z = 691.83 520 m/z = 664.76 (C50H33N3O = 691.26) (C47H26N4O = 664.22) 522 m/z = 841.97 524 m/z = 757.91 (C61H35N3O2 = 841.27) (C53H31N3OS = 757.21) 525 m/z = 757.91 527 m/z = 767.93 (C53H31N3OS = 757.21) (C56H37N3O = 767.29) 528 m/z = 740.86 531 m/z = 727.86 (C53H32N4O = 740.25) (C53H33N3O = 727.26) 532 m/z = 727.86 533 m/z = 803.96 (C53H33N3O = 727.26) (C59H37N3O = 803.29) 535 m/z = 727.86 536 m/z = 664.76 (C53H33N3O = 727.26) (C47H28N4O = 664.22) 539 m/z = 665.75 542 m/z = 640.86 (C47H27N3O2 = 665.21) (C53H32N4O = 740.25) 543 m/z = 691.83 544 m/z = 741.85 (C50H33N3O = 691.26) (C53H31N3O2 = 741.24) 547 m/z = 757.91 548 m/z = 807.97 (C53H31N3OS = 757.21) (C57H33N3OS = 807.23) 549 m/z = 816.96 550 m/z = 767.93 (C59H36N4O = 816.28) (C56H37N3O = 767.29) 554 m/z = 665.75 555 m/z = 728.85 (C46H26N4O = 652.22) (C52H32N4O = 728.25) 557 m/z = 728.85 558 m/z = 804.95 (C52H32N4O = 728.25) (C58H36N4O = 804.28) 564 m/z = 741.85 565 m/z = 692.82 (C52H31N5O = 741.25) (C49H32N4O = 692.26) 566 m/z = 665.75 569 m/z = 742.84 (C46H27N5O = 665.22) (C52H30N4O2 = 742.24) 573 m/z = 768.92 574 m/z = 741.85 (C55H36N4O = 768.29) (C52H31N5O = 741.25) 580 m/z = 521.62 564 m/z = 741.85 (C39H23NO = 521.18) (C52H31N5O = 741.25) 565 m/z = 692.82 566 m/z = 665.76 (C49H32N4O = 692.26) (C46H27N5O = 665.22) 569 m/z = 742.84 573 m/z = 768.92 (C52H30N4O2 = 742.24) (C55H36N4O = 768.29) 574 m/z = 741.85 580 m/z = 521.62 (C52H31N5O = 741.25) (C39H23NO = 521.18) 583 m/z = 834.01 584 m/z = 894.05 (C59H35N3OS = 833.25) (C65H39N3O2 = 893.30) 585 m/z = 599.69 587 m/z = 675.79 (C43H25N3O = 599.20) (C49H29N3O = 675.23) 588 m/z = 725.85 594 m/z = 662.79 (C53H31N3O = 725.25) (C49H30N2O = 662.24) 597 m/z = 652.76 598 m/z = 652.76 (C46H28N4O = 652.23) (C46H28N4O = 652.23) 599 m/z = 728.86 600 m/z = 728.86 (C52H32N4O = 728.26) (C52H32N4O = 728.26) 601 m/z = 666.74 605 m/z = 758.90 (C46H26H4O2 = 666.21) (C52H30N4OS = 758.21) 606 m/z = 741.85 611 m/z = 727.87 (C52H31N5O = 741.25) (C53H33N3O = 727.26) 612 m/z = 664.77 615 m/z = 740.87 (C47H28N4O = 664.23) (C53H32N4O = 740.26) 618 m/z = 767.93 620 m/z = 642.78 (C56H37N3O = 767.29) (C44H26N4S = 642.19) 623 m/z = 668.82 624 m/z = 794.98 (C46H28N4S = 668.20) (C56H34N4S = 794.25) 626 m/z = 821.01 629 m/z = 682.80 (C56H36N4S = 820.27) (C46H26N4OS = 682.18) 632 m/z = 698.86 634 m/z = 708.88 (C46H26N4S2 = 698.16) (C49H32N4S = 708.23) 635 m/z = 681.82 636 m/z = 758.90 (C46H27N5S = 681.20) (C52H30N4OS = 758.21) 640 m/z = 774.96 641 m/z = 834.01 (C52H30N4S2 = 774.19) (C58H35N5S = 833.26) 644 m/z = 587.74 646 m/z = 613.78 (C43H25NS = 587.17) (C45H27NS =

613.19) 649 m/z = 537.68 651 m/z = 715.91 (C39H23NS = 537.16) (C53H33NS = 715.23) 652 m/z = 669.86 653 m/z = 911.10 (C47H27NS2 = 669.16) (C64H38N4OS = 910.28) 654 m/z = 834.01 655 m/z = 768.98 (C59H35N3OS = 833.25) (C56H36N2S = 768.26) 657 m/z = 539.66 659 m/z = 741.91 (C37H21N3S = 539.15) (C53H31N3S = 741.22) 660 m/z = 668.82 661 m/z = 677.87 (C46H28N4S = 668.20) (C50H31NS = 677.22) 664 m/z = 652.82 665 m/z = 678.85 (C47H28N2S = 652.20) (C49H30N2S = 678.21) 666 m/z = 675.85 670 m/z = 676.84 (C50H29NS = 675.2) (C49H28N2S = 676.20) 672 m/z = 891.00 674 m/z = 668.82 (C58H34N4S = 818.25) (C46H28N4S = 668.2) 675 m/z = 744.92 677 m/z = 744.92 (C52H32N4S = 744.23) (C52H32N4S = 744.23) 680 m/z = 758.90 682 m/z = 758.90 (C52H30N4OS = 758.21) (C52H30N4OS = 758.21) 683 m/z = 849.04 685 m/z = 884.07 (C58H32N4S2 = 848.21) (C62H37N5S = 883.28) 687 m/z = 706.83 688 m/z = 758.90 (C47H26N6S = 706.19) (C52H30N4OS = 758.21) 691 m/z = 774.96 692 m/z = 875.08 (C52H30N4S2 = 774.19) (C60H34N4S2 = 874.22) 695 m/z = 757.91 696 m/z = 644.84 (C52H31N5S = 757.23) (C46H32H2S = 644.23) 697 m/z = 618.75 700 m/z = 768.94 (C43H26N2OS = 618.18) (C54H32N4S = 768.23) 703 m/z = 743.94 704 m/z = 834.01 (C53H29NS2 = 743.17) (C59H35N3OS = 833.25) 707 m/z = 691.85 708 m/z = 668.82 (C49H29N3S = 691.21) (C46H28N4S = 668.20) 709 m/z = 713.83 713 m/z = 667.83 (C49H32NOPS = 713.19) (C47H29N3S = 667.21) 714 m/z = 743.93 716 m/z = 743.93 (C53H33N3S = 743.24) (C53H33N3S = 743.24) 717 m/z = 820.03 721 m/z = 793.99 (C59H37N3S = 819.27) (C57H35N3S = 793.26) 728 m/z = 808.01 729 m/z = 680.83 (C58H37N3S = 807.27) (C47H28N4S = 680.20) 731 m/z = 757.91 732 m/z = 757.91 (C53H31N3OS = 757.22) (C53H31N3OS = 757.22) 733 m/z = 824.03 734 m/z = 824.03 (C57H33N3S2 = 823.21) (C57H33N3S2 = 823.21) 735 m/z = 833.03 736 m/z = 783.99 (C59H36N4S = 832.27) (C56H37N3S = 783.27) 741 m/z = 820.03 743 m/z = 793.99 (C59H37N3S = 819.27) (C57H35N3S = 793.26) 745 m/z = 743.93 748 m/z = 665.75 (C53H33N3S = 743.24) (C47H27N3O2 = 664.21) 750 m/z = 747.93 752 m/z = 707.89 (C51H29N3S2 = 747.18) (C50H33N3S = 707.24) 756 m/z = 757.91 757 m/z = 824.03 (C53H31N3OS = 757.22) (C57H33N3S2 = 823.21) 761 m/z = 756.93 764 m/z = 768.94 (C53H32N4S = 756.23) (C54H32N4S = 768.23) 766 m/z = 793.99 767 m/z = 591.73 (C57H35N3S = 793.26) (C41H25N3S = 591.18) 769 m/z = 718.88 771 m/z = 668.82 (C50H30N4S = 718.22) (C46H28N4S = 668.20) 773 m/z = 744.92 775 m/z = 845.04 (C52H32N4S = 744.23) (C60H36N4S = 844.27) 779 m/z = 782.92 783 m/z = 698.86 (C54H30N4OS = 782.21) (C46H26N4S2 = 698.16) 784 m/z = 757.91 786 m/z = 781.94 (C52H31N5S = 757.23) (C54H31N5S = 781.23) 787 m/z = 758.90 789 m/z = 758.90 (C52H30N4OS = 758.21) (C52H30N4OS = 758.21) 791 m/z = 774.96 792 m/z = 834.01 (C52H30N4S2 = 774.19) (C58H35N5S = 833.26) 793 m/z = 885.10 798 m/z = 589.76 (C63H40N4S = 884.30) (C43H27NS = 589.19) 801 m/z = 644.84 802 m/z = 537.68 (C46H32N2S = 644.23) (C39H23NS = 537.16) 807 m/z = 758.90 809 m/z = 615.75 (C52H30N4OS = 758.21) (C43H25N3S = 615.18) 810 m/z = 539.66 814 m/z = 675.85 (C37H21N3S = 539.15) (C50H29NS = 675.20) 816 m/z = 641.79 817 m/z = 565.69 (C4527N3S = 641.19) (C39H23N3S = 565.16) 818 m/z = 602.75 822 m/z = 845.04 (C43H26N2S = 602.18) (C60H36N4S = 844.27) 823 m/z = 847.05 825 m/z = 742.90 (C60H38N4S = 846.28) (C52H30N4S = 742.22) 829 m/z = 744.92 831 m/z = 821.01 (C52H32N4S = 744.23) (C58H36N4S = 820.27) 833 m/z = 758.90 836 m/z = 748.92 (C52H30N4OS = 758.21) (C50H28N4S2 = 748.18) 838 m/z = 884.07 842 m/z = 758.90 (C62H37N5S = 883.28) (C52H30N4OS = 758.21) 844 m/z = 774.96 847 m/z = 784.98 (C52H304S2 = 774.19) (C55H36N4S = 784.27) 848 m/z = 757.91 849 m/z = 644.84 (C52H31N5S = 757.23) (C46H32N2S = 644.23) 858 m/z = 835.00 860 m/z = 691.85 (C58H34N4OS = 834.25) (C49H29N3S = 691.21) 861 m/z = 668.82 862 m/z = 713.83 (C46H28N4S = 668.20) (C49H32OPS = 713.19) 863 m/z = 678.85 870 m/z = 667.83 (C49H30N2S = 679.21) (C47H29N3S = 667.21) 871 m/z = 743.93 872 m/z = 820.03 (C53H33N3S = 743.24) (C59H37N3S = 819.27) 883 m/z = 757.91 884 m/z = 757.91 (C53H31N3OS = 757.22) (C53H31N3OS = 757.22) 887 m/z = 833.03 888 m/z = 783.99 (C59H36N4S = 832.27) (C56H37N3S = 783.27) 890 m/z = 591.73 893 m/z = 820.03 (C41H25N3S

= 591.18) (C59H37N3S = 819.27) 905 m/z = 757.91 908 m/z = 773.97 (C53H31N3OS = 757.22) (C53H31N3S2 = 773.20) 910 m/z = 783.99 911 m/z = 756.93 (C56H37N3S = 783.27) (C53H32N4S = 756.23) 916 m/z = 744.92 918 m/z = 821.01 (C52H32N4S = 744.23) (C58H36N4S = 820.27) 919 m/z = 682.80 920 m/z = 682.80 (C46H26N4OS = 682.18) (C46H26N4OS = 682.18) 923 m/z = 698.86 924 m/z = 757.91 (C46H26N4S2 = 698.16) (C52H31N5S = 757.23) 930 m/z = 774.96 931 m/z = 774.96 (C52H30N4S2 = 774.19) (C52H30N4S2 = 774.19) 932 m/z = 834.01 943 m/z = 834.01 (C58H35N5S = 833.26) (C59H35N3OS = 833.25) 945 m/z = 539.66 952 m/z = 564.71 (C37H21N3S = 539.15) (C40H24N2S = 563.17) 956 m/z = 678.85 957 m/z = 675.85 (C49H30N2S = 678.21) (C50H29NS = 675.20) 960 m/z = 600.74 963 m/z = 768.94 (C43H24N2S = 600.17) (C54H32N4S = 768.23) 965 m/z = 668.82 968 m/z = 821.01 (C46H28N4S = 668.20) (C58H36N4S = 820.27) 971 m/z = 758.90 979 m/z = 758.90 (C52H30N4OS = 758.21) (C52H30N4OS = 758.21) 980 m/z = 758.90 984 m/z = 834.01 (C52H30N4OS = 758.21) (C58H35H5S = 833.26) 986 m/z = 757.91 991 m/z = 613.78 (C52H31N5S = 757.23) (C45H27NS = 613.19) 998 m/z = 668.82 995 m/z = 834.01 (C46H28N4S = 668.20) (C59H35N3OS = 833.25) 1000 m/z = 678.85 1004 m/z = 743.93 (C49H30N2S = 678.21) (C53H33N3S = 743.24) 1006 m/z = 667.83 1011 m/z = 681.81 (C47H29N3S = 667.21) (C47H27N3SO = 681.19) 1015 m/z = 707.89 1080 m/z = 835.00 (C50H33N3S = 707.24) (C58H34N4OS = 834.25) 1086 m/z = 691.85 1098 m/z = 676.64 (C49H29N3S = 691.21) (C49H28N2S = 676.20) 1099 m/z = 692.84 (C48H26N4S = 692.20)

#### <Experimental Example 1> Manufacture of Organic Light Emitting Device

##### Comparative Example 1

##### (172) 1) Manufacture of Organic Light Emitting Device

(173) A transparent indium tin oxide (ITO) electrode thin film obtained from glass for an OLED (manufactured by Samsung-Corning Co., Ltd.) was ultrasonic cleaned using trichloroethylene, acetone, ethanol and distilled water consecutively for 5 minutes each, stored in isopropanol, and used.

(174) Next, an ITO substrate was installed in a substrate folder of a vacuum deposition apparatus, and the following 4,4',4''-tris(N,N-(2-naphthyl)-phenylamino)triphenylamine (2-TNATA) was introduced to a cell in the vacuum deposition apparatus.

(175) ##STR01068##

(176) Subsequently, the chamber was evacuated until the degree of vacuum therein reached 10.sup.-6 torr, and then 2-TNATA was evaporated by applying a current to the cell to deposit a hole injection layer having a thickness of 600 Å on the ITO substrate.

(177) To another cell of the vacuum deposition apparatus, the following N,N'-bis(α-naphthyl)-N,N'-diphenyl-4,4'-diamine (NPB) was introduced, and evaporated by applying a current to the cell to deposit a hole transfer layer having a thickness of 300 Å on the hole injection layer.

(178) ##STR01069##

(179) After forming the hole injection layer and the hole transfer layer as above, a blue light emitting material having a structure as below was deposited thereon as a light emitting layer. Specifically, in one side cell in the vacuum deposition apparatus, H1, a blue light emitting host material, was vacuum deposited to a thickness of 200 Å, and D1, a blue light emitting dopant material, was vacuum deposited thereon by 5% with respect to the host material.

(180) ##STR01070##

(181) Subsequently, a compound of the following Structural Formula E1 was deposited to a thickness of 300 Å as an electron transfer layer.

(182) ##STR01071##

(183) As an electron injection layer, lithium fluoride (LiF) was deposited to a thickness of 10 Å, and an Al cathode was employed to a thickness of 1,000 Å, and as a result, an OLED was manufactured.

(184) Meanwhile, all the organic compounds required to manufacture the OLED were vacuum

sublimation purified under 10.sup.-8 torr to 10.sup.-6 torr by each material to be used in the OLED manufacture.

(185) ##STR01072##

Comparative Example 2 to Comparative Example 5

(186) Organic electroluminescent devices were manufactured in the same manner as in Comparative Example 1 except that Compound A, Compound B, Compound C and Compound D were used instead of E1 used when forming the electron transfer layer.

Example 1 to Example 469

(187) Organic electroluminescent devices were manufactured in the same manner as in Comparative Example 1 except that compounds listed in the following Table 20 were used instead of E1 used when forming the electron transfer layer.

(188) Results of measuring driving voltage, light emission efficiency, color coordinate (CIE) and lifetime of each of the blue organic light emitting devices manufactured according to the present disclosure are as shown in the following Table 20.

(189) TABLE-US-00020 TABLE 20 Light Driving Emission Voltage Efficiency CIE Lifetime  
Compound (V) (cd/A) (x, y) (T95) Comparative E1 5.21 4.11 (0.134, 24 Example 1 0.100)  
Comparative A 4.13 3.20 (0.134, 13 Example 2 0.104) Comparative B 4.11 3.12 (0.134, 10  
Example 3 0.104) Comparative C 5.67 5.11 (0.134, 21 Example 4 0.104) Comparative D 5.67 5.12  
(0.134, 20 Example 5 0.104) Example 1 1 5.44 6.47 (0.134, 36 0.102) Example 2 4 4.47 6.87  
(0.134, 40 0.100) Example 3 5 4.67 6.64 (0.129, 38 0.100) Example 4 7 4.82 6.55 (0.130, 36 0.099)  
Example 5 8 4.44 6.97 (0.134, 40 0.101) Example 6 9 4.61 6.89 (0.134, 40 0.103) Example 7 11  
5.35 6.30 (0.134, 33 0.102) Example 8 12 5.62 5.95 (0.134, 42 0.103) Example 9 13 5.40 6.12  
(0.134, 39 0.101) Example 10 15 5.50 5.89 (0.134, 41 0.100) Example 11 16 5.44 6.01 (0.134, 36  
0.101) Example 12 17 5.34 6.58 (0.134, 45 0.100) Example 13 18 5.38 6.93 (0.134, 43 0.100)  
Example 14 24 4.91 6.32 (0.134, 41 0.100) Example 15 25 4.98 6.44 (0.134, 40 0.100) Example 16  
28 5.62 6.38 (0.134, 35 0.100) Example 17 30 5.66 6.04 (0.131, 30 0.102) Example 18 35 5.40  
6.49 (0.134, 31 0.101) Example 19 36 5.60 6.22 (0.129, 29 0.100) Example 20 37 4.70 6.01 (0.134,  
36 0.101) Example 21 38 5.40 6.12 (0.134, 44 0.103) Example 22 41 5.60 6.21 (0.134, 43 0.102)  
Example 23 42 5.45 6.22 (0.134, 37 0.101) Example 24 44 5.39 5.95 (0.134, 33 0.102) Example 25  
45 4.96 5.95 (0.134, 42 0.101) Example 26 47 4.91 6.13 (0.134, 39 0.101) Example 27 52 4.91  
5.85 (0.134, 41 0.100) Example 28 53 4.98 6.38 (0.134, 42 0.101) Example 29 56 4.91 5.85 (0.134,  
41 0.100) Example 30 57 4.98 6.38 (0.134, 42 0.101) Example 31 58 5.44 6.47 (0.134, 36 0.102)  
Example 32 64 4.47 6.87 (0.134, 40 0.100) Example 33 68 4.67 6.64 (0.129, 38 0.100) Example 34  
69 4.82 6.55 (0.130, 36 0.099) Example 35 70 4.44 6.97 (0.134, 40 0.101) Example 36 72 4.61  
6.89 (0.134, 40 0.103) Example 37 75 5.35 6.30 (0.134, 33 0.102) Example 38 77 5.62 5.95 (0.134,  
42 0.103) Example 39 79 5.40 6.12 (0.134, 39 0.101) Example 40 80 5.50 5.89 (0.134, 41 0.100)  
Example 41 84 5.44 6.01 (0.134, 36 0.101) Example 42 86 5.34 6.58 (0.134, 45 0.100) Example 43  
88 5.38 6.93 (0.134, 43 0.100) Example 44 89 4.91 6.32 (0.134, 41 0.100) Example 45 91 4.98  
6.44 (0.134, 40 0.100) Example 46 92 5.62 6.38 (0.134, 35 0.100) Example 47 97 5.66 6.04 (0.131,  
30 0.102) Example 48 100 5.40 6.49 (0.134, 31 0.101) Example 49 102 5.60 6.22 (0.129, 29 0.100)  
Example 50 103 4.70 6.01 (0.134, 36 0.101) Example 51 107 5.40 6.12 (0.134, 44 0.103) Example  
52 112 5.60 6.21 (0.134, 43 0.102) Example 53 113 5.45 6.22 (0.134, 37 0.101) Example 54 114  
5.39 5.95 (0.134, 33 0.102) Example 55 116 4.96 5.95 (0.134, 42 0.101) Example 56 119 5.44 6.47  
(0.134, 36 0.102) Example 57 120 4.47 6.87 (0.134, 40 0.100) Example 58 121 4.67 6.64 (0.129,  
38 0.100) Example 59 128 4.82 6.55 (0.130, 36 0.099) Example 60 132 4.44 6.97 (0.134, 40 0.101)  
Example 61 133 4.61 6.89 (0.134, 40 0.103) Example 62 135 5.35 6.30 (0.134, 33 0.102) Example  
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Example 451 965 5.62 5.98 (0.134, 33 0.100) Example 452 968 4.91 6.32 (0.134, 41 0.100)  
Example 453 971 4.98 6.44 (0.134, 40 0.100) Example 454 979 5.62 6.38 (0.134, 35 0.100)  
Example 455 980 5.44 6.34 (0.134, 36 0.102) Example 456 984 5.62 6.20 (0.134, 39 0.101)  
Example 457 986 5.62 6.22 (0.134, 47 0.100) Example 458 991 5.62 5.98 (0.134, 33 0.100)  
Example 459 995 4.72 6.55 (0.134, 48 0.102) Example 460 998 4.72 6.20 (0.134, 43 0.102)  
Example 461 1000 5.40 6.12 (0.134, 39 0.101) Example 462 1004 5.44 6.21 (0.134, 41 0.100)  
Example 463 1006 5.39 6.20 (0.134, 36 0.101) Example 464 1011 4.96 6.88 (0.134, 45 0.100)  
Example 465 1015 4.91 6.93 (0.134, 43 0.100) Example 466 1080 4.96 6.88 (0.134, 45 0.100)  
Example 467 1086 4.91 6.93 (0.134, 43 0.102) Example 468 1098 4.98 6.22 (0.134, 40 0.100)  
Example 469 1099 4.72 6.53 (0.134, 48 0.102)

(190) As seen from Table 20, the organic light emitting device using the compound of the present disclosure as an electron transfer layer material of a blue organic light emitting device had lower

driving voltage and improved light emission efficiency and lifetime compared to Comparative Example 1, Comparative Example 2, Comparative Example 3 and Comparative Example 4. Such results are due to the fact that the compound is a bipolar type having both a p-type and an n-type, and therefore, hole leakage is prevented and electrons are effectively injected to a light emitting layer.

## Claims

1. A heterocyclic compound represented by the following Chemical Formula 1: ##STR01073## wherein, in Chemical Formula 1, X is O or S, L is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, Z is a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group; or a substituted or unsubstituted phosphine oxide group, R1 is a substituted or unsubstituted C6 to C30 aryl group, R2 to R7 are each independently hydrogen; or deuterium, a is 1 or 2, b is an integer of 1 to 3, m and n are each an integer of 1 to 5, and when a, b, m and n are each 2 or greater, substituents in the parentheses are the same as or different from each other.
2. The heterocyclic compound of claim 1, wherein Chemical Formula 1 is represented by any one of the following Chemical Formula 2 to Chemical Formula 4: ##STR01074## in Chemical Formulae 2 to 4, X, L, Z, R1 to R7, a, b, m and n have the same definitions as in Chemical Formula 1.
3. The heterocyclic compound of claim 1, wherein L is a direct bond; a substituted or unsubstituted phenylene group; a substituted or unsubstituted biphenylene group; a substituted or unsubstituted terphenylene group; a substituted or unsubstituted naphthylene group; a substituted or unsubstituted anthracenylene group; a substituted or unsubstituted triphenylenylene group; a substituted or unsubstituted phenanthrenylene group; a substituted or unsubstituted divalent pyridine group; a substituted or unsubstituted divalent pyrimidine group; or a substituted or unsubstituted divalent triazine group.
4. The heterocyclic compound of claim 1, wherein Z is a substituted or unsubstituted phenyl group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a substituted or unsubstituted pyridine group; a substituted or unsubstituted pyrimidine group; a substituted or unsubstituted triazine group; a substituted or unsubstituted carbazole group; a substituted or unsubstituted benzocarbazole group; a substituted or unsubstituted phenanthrolyl group; a substituted or unsubstituted dibenzofuran group; a substituted or unsubstituted dibenzothiophene group; a substituted or unsubstituted dihydroacridine group; or a substituted or unsubstituted phosphine oxide group.
5. The heterocyclic compound of claim 1, wherein Chemical Formula 1 is represented by any one of the following compounds: ##STR01075## ##STR01076## ##STR01077## ##STR01078## ##STR01079## ##STR01080## ##STR01081## ##STR01082## ##STR01083## ##STR01084## ##STR01085## ##STR01086## ##STR01087## ##STR01088## ##STR01089## ##STR01090## ##STR01091## ##STR01092## ##STR01093## ##STR01094## ##STR01095## ##STR01096## ##STR01097## ##STR01098## ##STR01099## ##STR01100## ##STR01101## ##STR01102## ##STR01103## ##STR01104## ##STR01105## ##STR01106## ##STR01107## ##STR01108## ##STR01109## ##STR01110## ##STR01111## ##STR01112## ##STR01113## ##STR01114## ##STR01115## ##STR01116## ##STR01117## ##STR01118## ##STR01119## ##STR01120## ##STR01121## ##STR01122## ##STR01123## ##STR01124## ##STR01125## ##STR01126## ##STR01127## ##STR01128## ##STR01129## ##STR01130## ##STR01131## ##STR01132## ##STR01133## ##STR01134## ##STR01135## ##STR01136## ##STR01137## ##STR01138## ##STR01139## ##STR01140## ##STR01141## ##STR01142## ##STR01143## ##STR01144##

##STR01145## ##STR01146## ##STR01147## ##STR01148## ##STR01149## ##STR01150##  
##STR01151## ##STR01152## ##STR01153## ##STR01154## ##STR01155## ##STR01156##  
##STR01157## ##STR01158## ##STR01159## ##STR01160## ##STR01161## ##STR01162##  
##STR01163## ##STR01164## ##STR01165## ##STR01166## ##STR01167## ##STR01168##  
##STR01169## ##STR01170## ##STR01171## ##STR01172## ##STR01173## ##STR01174##  
##STR01175## ##STR01176## ##STR01177## ##STR01178## ##STR01179## ##STR01180##  
##STR01181## ##STR01182## ##STR01183## ##STR01184## ##STR01185## ##STR01186##  
##STR01187## ##STR01188## ##STR01189## ##STR01190## ##STR01191## ##STR01192##  
##STR01193## ##STR01194## ##STR01195## ##STR01196## ##STR01197## ##STR01198##  
##STR01199## ##STR01200## ##STR01201## ##STR01202## ##STR01203## ##STR01204##  
##STR01205## ##STR01206## ##STR01207## ##STR01208## ##STR01209## ##STR01210##  
##STR01211## ##STR01212## ##STR01213## ##STR01214## ##STR01215## ##STR01216##  
##STR01217## ##STR01218## ##STR01219## ##STR01220## ##STR01221## ##STR01222##  
##STR01223## ##STR01224## ##STR01225## ##STR01226## ##STR01227## ##STR01228##  
##STR01229## ##STR01230## ##STR01231## ##STR01232## ##STR01233## ##STR01234##  
##STR01235## ##STR01236## ##STR01237## ##STR01238## ##STR01239## ##STR01240##  
##STR01241## ##STR01242## ##STR01243## ##STR01244## ##STR01245## ##STR01246##  
##STR01247## ##STR01248## ##STR01249## ##STR01250## ##STR01251## ##STR01252##  
##STR01253## ##STR01254## ##STR01255## ##STR01256## ##STR01257## ##STR01258##  
##STR01259## ##STR01260## ##STR01261## ##STR01262## ##STR01263## ##STR01264##  
##STR01265## ##STR01266## ##STR01267## ##STR01268## ##STR01269## ##STR01270##  
##STR01271## ##STR01272##

6. An organic light emitting device comprising: a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes the heterocyclic compound of claim 1.

7. The organic light emitting device of claim 6, wherein the organic material layer includes an electron transfer layer, and the electron transfer layer includes the heterocyclic compound.

8. The organic light emitting device of claim 6, further comprising one, two or more layers selected from the group consisting of a light emitting layer, a hole injection layer, a hole transfer layer, an electron injection layer, an electron transfer layer, an electron blocking layer and a hole blocking layer.

9. The organic light emitting device of claim 6 comprising: the first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and the second electrode provided on the second stack.

10. The organic light emitting device of claim 9, wherein the charge generation layer includes the heterocyclic compound.

11. The organic light emitting device of claim 9, wherein the charge generation layer is an N-type charge generation layer, and the charge generation layer includes the heterocyclic compound.

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